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Macroeconomic Drivers of Inflation Expectations and Inflation Risk Premia

Jef Boeckx[1], Leonardo Iania [2,3], and Joris Wauters [1]

[1]Economics and Research Department, National Bank of Belgium, de Berlaimontlaan 14, Brussels, 1000, Belgium
[2]CORE, UCLouvain, Voie du Roman Pays 34, Louvain la Neuve, 1348, Belgium
[3]Faculty of Economics and Business, KULeuven, Naamsestraat 69, Leuven, 3000, Belgium

Address correspondence to CORE, UCLouvain, Voie du Roman Pays 34, Louvain la Neuve, 1348, Belgium, or email: leonardo.iania@uclouvain.be.

Abstract

We propose a new affine term structure model to decompose inflation-linked swap (ILS) rates into genuine inflation expectations and inflation risk premia. The model accounts for both short-term macroeconomic factors and long-term economic trends, namely trend inflation and the equilibrium real interest rate. We estimate the model for the US and euro area and find that the inflation risk premium significantly drives ILS rates, with macroeconomic shocks playing different roles across maturities and regions. International factors, particularly oil prices, impact market-based inflation expectations during key global events like the COVID-19 pandemic and the Russian invasion of Ukraine.

Keywords: Inflation-linked swaps, affine term structure model, inflation expectations, inflation risk premia, trend inflation, shifting endpoints

JEL classifications: E31, E44, E52

Central banks pay great attention to analyzing the behavior of inflation expectations, as they are an important determinant of the inflation outlook. When households, firms, or financial markets expect future inflation to be misaligned with the central bank's target, risks are large of a self-fulfilling prophecy if the central bank does not respond adequately.

However, gauging inflation expectations is challenging because available measures present strengths and weaknesses. For example, surveys conducted with households, firms, or professional economists directly measure the private sector's inflation expectations, but they are conducted infrequently and may depend on specific survey design. Financial markets offer another source of information on inflation expectations: inflation-linked swaps (ILS), whose payouts are directly linked to the level of inflation and available for a wide range of maturities, have become an increasingly liquid asset class. Their availability at a very high frequency and the fact that traders have "skin in the game" when trading

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inflation products are obvious advantages of this information source. But market participants require compensation for the uncertain returns of investing in these products: their prices also contain risk premia, on top of the "pure" expectations of future inflation. That necessitates using methods to estimate these risk premia and better understand their behavior.

This article proposes a new affine term structure model (ATSM) to decompose ILS rates as the sum of a genuine inflation expectations component (EC) and an inflation risk premium (IRP). Our approach innovates by measuring how macroeconomic drivers contribute to their dynamics. Specifically, we separate the macroeconomic drivers into transitory (or short-run) parts and long-term, economically grounded determinants. The latter contains time-varying trend inflation and equilibrium real interest rates. Both have been trending down since the mid-80s but have shown upward movement in recent years. Disregarding the pronounced low-frequency movements in these variables risks overestimating how strongly the IRP determines the movements of ILS rates.

Our model follows Joslin, Singleton, and Zhu (2011), in that, under the risk-neutral measure, ILS rates follow a low-dimensional factor structure with spanned factors. The historical evolution of ILS rates, latent factors, macroeconomic variables, and long-run trends has richer dynamics: it obeys a higher-order vector error correction model with theory-grounded cointegration relationships. Hence, ILS rates and the IRP depend on (i) latent variables, namely the principal components of the ILS term structure; (ii) local macroeconomic variables, that is, the short-run interest rate, inflation, and real activity; (iii) international energy (oil) prices and global economic activity; and (iv) slow-moving variables, that is, trend inflation and the equilibrium real interest rate, that capture low-frequency movements in inflation and the interest rate.

We estimate the model separately for the United States (US) and euro area (EA). Our state-space setting exploits mixed-frequency sources of information to identify the long-run trends, such as monthly macroeconomic data and quarterly survey data from the Survey of Professional Forecasters and the monthly/bi-monthly and quarterly data from Consensus Economics surveys. To ease the estimation burden and avoid over-fitting problems linked to many model parameters, we apply a penalized likelihood approach that shrinks high-order lag parameters (see, e.g. Tibshirani 1996; Fan and Li 2001).

We present three main results. First, the IRP is an important driver of ILS rates. Our estimated IRP for the US is more volatile than in Fleckenstein, Longstaff, and Lustig (2017) and explains most of the variation in longer-term ILS rates for both regions. At the (short-term) 2-year maturity, the IRP was positive before the Great Financial Crisis (GFC) in both regions but then switched signs and became negative for a prolonged period in the EA. Therefore, a careful reading of inflation swaps is needed to conclude whether inflation expectations are anchored.

Second, a full-sample variance decomposition reveals that the relative importance of short- and long-term innovations in driving ILS rates depends on the contract's maturity and the movement frequency. For instance, 2-year ILS contracts are significantly driven by transitory macroeconomic developments in both regions. However, while international factors (oil prices and global activity) are essential in the US, the weight is tilted toward local factors (i.e. inflation, interest rates, industrial production) in the EA. When considering 10-year contracts, we find for both regions that long-term innovations (trend inflation and natural rate) play a larger role in explaining longer-term ILS rates compared to 2-year contracts. This result applies particularly to low-frequency (i.e. 10-year horizon) movements in the EC, which are dominantly explained by such shocks. Finally, spanned factor shocks to the principal components of the ILS term structure are crucial to IRP variations, accounting for roughly one-fifth (half) of the US (EA) high- and low-frequency IRP movements.

Third, zooming in on four recent episodes—the GFC, the oil price glut, COVID-19, and the Russian invasion of Ukraine—we find that international factors were crucial drivers, particularly oil prices. For 2-year contracts, these drivers work through the EC and IRP channels, implying that they are incorporated in market-based inflation expectations, even when refined from risk compensation.

1.1 Related literature and contributions

This article contributes to the macro-finance literature that applies term structure models to disentangle inflation expectations from risk premia in market-based measures. Such models have been mainly applied to bond market yields (see, e.g. Ang, Bekaert, and Wei 2008; Christensen, Lopez, and Rudebusch 2010; Joyce, Lildholdt, and Sorensen 2010; Hördahl and Tristani 2014; Berardi and Plazzi 2019) and example extensions account for liquidity premia in inflation-linked government bonds (Abrahams et al. 2016) or the zero lower bound in a consistent shadow rate model (Carriero, Mouabbi, and Vangelista 2018).

Haubrich, Pennacchi, and Ritchken (2012) introduce ILS rates in term structure estimation for the US, and Fleckenstein, Longstaff, and Lustig (2017) use market prices of inflation swaps and options to measure deflation risk.[1] Estimates of inflation risk premia and expectations using ILS rates are given by Camba-Mendez and Werner (2017) for the EA and US, and by Cecchetti, Grasso, and Pericoli (2022) and Neri et al. (2022) for the EA.[2]

Our article departs from the above studies in two ways. First, our *methodological* contribution is that we allow for time-variation in trend inflation and the equilibrium real interest rate in the decomposition of ILS rates. We build on Dewachter and Iania (2011), who propose a Macro-Finance model of the US nominal yield curve including macroeconomic, liquidity-related, and return forecasting factors, and Bauer and Rudebusch (2020), who extend the normalization setting of Joslin, Singleton, and Zhu (2011) to account for stochastic trends in inflation and the real interest rate. To identify these trend components, we include survey data on inflation expectations and natural rate estimates of Holston, Laubach, and Williams (2017) into the model. We find that disregarding these variables' pronounced low-frequency movements has implications for the final estimates. Specifically, by decomposing IRP and EC components in deterministic and stochastic parts (see Giannone, Lenza, and Primiceri 2019), we show that a latent factor model for the EA, which excludes long-term trends, generates oscillating behavior in the deterministic part of the EC. This unrealistic feature is absent in our model.

Second, our *empirical* contribution is that, by integrating both short- versus long-term and local versus global shocks, we offer a more comprehensive understanding of the dynamics influencing ILS rates and their subcomponents in the US and EA. Moreover, our sample is broader by covering the recent COVID-19 pandemic and the inflation outburst following the Russian invasion.

In rest of the article, we outline our model in Section 2, present the dataset and the estimation strategy in Section 3, report and discuss our results in Section 4, and finally, we draw the concluding remarks in Section 5. Appendix A reports additional results and parameter estimates.

2. Modeling Approach

2.1 Term Structure Modeling

2.1.1 Defining the ILS contracts

A zero-coupon ILS is a contract by which, at maturity, the floating leg counterpart receives a payment linked to the realized inflation rate throughout the contract (n), $\pi_{t+n,t} = \frac{P_{t+n}}{P_t} - 1$. In exchange, this party pays an amount linked to a fixed rate, $Y_{n,t}$, established at the

[1] Berardi and Plazzi (2019) report that their findings are robust to replacing US TIPS yields by ILS rates.

[2] Höynck and Rossi (2023) identify the contribution from different shocks on US and EA ILS rates using a Bayesian VAR model with sign and magnitude restrictions. However, their approach does not separate ILS rates into expectations and risk premia components.

initiation of the contract.[3] Since there is no exchange of money at the initiation of the contract, the expected value of the two payments should be equal:

$$\underbrace{\mathbb{E}_t^p\left[M_{t,t+n}[(1+Y_{n,t})^n - 1]\right]}_{\text{Fixed leg}} = \underbrace{\mathbb{E}_t^p[M_{t,t+n}\pi_{t+n,t}]}_{\text{Floating leg}} \quad p = \mathbb{P}, \mathbb{Q}, \tag{1}$$

where $M_{t,t+n}$ is the stochastic discount factor. The equality reported in Equation (1) is valid in probability settings adjusted for risk, namely risk-neutral probability measures $\mathbb{Q}$, and under the historical probability measures $\mathbb{P}$. As shown by Camba-Mendez and Werner (2017), pricing the contract under the risk-neutral measure simplifies Equation (1) as:

$$y_{n,t} = \frac{1}{n}\log(\mathbb{E}_t^{\mathbb{Q}} e^{\sum_{j=1}^{n} \pi_{t+j}}), \tag{2}$$

where $y_{n,t}$ is the continuous time counterpart of $Y_{n,t}$ and π_{t+j} is the inflation rate between two consecutive periods, $t+j\text{-}1$ and $t+j$. Equation (2) highlights that, in the risk-neutral probabilistic setting, the (continuously compounded) ILS rate is the expected average one-period inflation rate throughout the contract.

2.1.2 Accounting for risk

We analyze the risk compensation in ILS contracts by focusing on the IRP obtained via the decomposition of ILS in expected (EC) and risk premium (IRP) components, respectively. We use our dynamic term structure modeling setting to decompose $y_{n,t}$ in EC and IRP components:

$$y_{n,t} = y_{n,t}^{EC} + y_{n,t}^{IRP}, \tag{3}$$

where $y_{n,t}^{EC}$ is the average expected inflation free of the IRP, and $y_{n,t}^{IRP}$ represents the compensation required by market participants for their exposure to the model's risk factors. We obtain the decomposition in Equation (3) in two steps. First, following Camba-Mendez and Werner (2017), we employ a Gaussian affine no-arbitrage setting to price the ILS contracts. ILS rates are expressed as an affine function of the risk factors $\mathcal{Z}_t$, identified in Section 2.2, by setting (i) the one-period ILS rate, $y_{1,t}$, an affine function of $\mathcal{Z}_t$, and (ii) by assuming that the risk factors have affine Gaussian dynamics under both probability measures:

$$y_{1,t} = \rho_0 + \boldsymbol{\rho}_1' \mathcal{Z}_t \tag{4a}$$

$$\mathcal{Z}_t = \boldsymbol{\mu}^i + \boldsymbol{\Phi}^i \mathcal{Z}_{t-1} + \boldsymbol{\Sigma}\boldsymbol{\varepsilon}_t^i, \quad \boldsymbol{\varepsilon}_t^i \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}) \text{ and } i = \mathbb{Q}, \mathbb{P}, \tag{4b}$$

where ρ_0, $\boldsymbol{\rho}_1$, $\boldsymbol{\mu}^{\mathbb{P}}$, $\boldsymbol{\mu}^{\mathbb{Q}}$, $\boldsymbol{\Phi}^{\mathbb{P}}$, $\boldsymbol{\Phi}^{\mathbb{Q}}$, and $\boldsymbol{\Sigma}$ are (potential) model parameters while $\boldsymbol{I}$ is an identity matrix of conformable dimension. The parameters of Equation (4b) under $\mathbb{Q}$ determine the pricing of the cross-sections of ILS rates, while the $\mathbb{P}$ parameters are key for the risk/term premium evolution. In our setting, the link between the two measures is obtained by assuming (i) essentially affine market prices of risk, $\boldsymbol{\Lambda}_t$, and (ii) a stochastic discount factor exponentially affine on the short-term ILS rate and $\boldsymbol{\Lambda}_t$:

[3] The exchange of payments is performed at maturity, and the exact quantity of cash to exchange is proportional to the contract's notional amount, multiplied by the difference between the realized inflation rate and the swap rate.

$$\boldsymbol{\Lambda}_t = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\lambda}_0 + \boldsymbol{\lambda}_1 \mathcal{Z}_t), \quad \boldsymbol{\mu}^{\mathbb{Q}} = \boldsymbol{\mu}^{\mathbb{P}} - \boldsymbol{\lambda}_0, \quad \boldsymbol{\Phi}^{\mathbb{Q}} = \boldsymbol{\Phi}^{\mathbb{P}} - \boldsymbol{\lambda}_1 \tag{5a}$$

$$M_{t,t+1} = \exp(-y_{1,t} - \frac{1}{2}\boldsymbol{\Lambda}_t'\boldsymbol{\Lambda}_t - \boldsymbol{\Lambda}_t'\boldsymbol{\Sigma}\boldsymbol{\varepsilon}_t^{\mathbb{P}}). \tag{5b}$$

Under these assumptions, it is possible to express $y_{n,t}$ as an affine function of the risk factors, $\mathcal{Z}_t$:

$$y_{n,t} = a_n + \boldsymbol{b}_n \mathcal{Z}_t, \tag{6}$$

whereby the weights of Equation (6) derive from the standard Riccati equations (see Ang and Piazzesi 2003). Specifically, $a_n = -n^{-1}a_n$ and $\boldsymbol{b}_n = -n^{-1}b_n$

$$a_n = a_{n-1} + b_{n-1}\boldsymbol{\mu}^{\mathbb{Q}} + \frac{1}{2}b_{n-1}\boldsymbol{\Sigma}\boldsymbol{\Sigma}'b_{n-1} - \rho_0 \tag{7a}$$

$$b_n = \boldsymbol{\Phi}^{\mathbb{Q}'} b_{n-1} - \boldsymbol{\rho}_1. \tag{7b}$$

Second, to obtain an expression for the EC of $y_{n,t}$, that is, the part of the ILS rates net from the risk compensations, we first switch off risk compensation by setting $\boldsymbol{\lambda}_0$ and $\boldsymbol{\lambda}_1$ to zero in Equation (5a). Subsequently, we derive a new set of loadings, a_n^{EC} and $\boldsymbol{b}_n^{EC}$, by iterating the Riccati equations with $\boldsymbol{\mu}^{\mathbb{P}}$ and $\boldsymbol{\Phi}^{\mathbb{P}}$ instead of $\boldsymbol{\mu}^{\mathbb{Q}}$ and $\boldsymbol{\Phi}^{\mathbb{Q}}$. As a result, the EC of $y_{n,t}$ is:

$$y_{n,t}^{EC} = a_n^{EC} + \boldsymbol{b}_n^{EC}\mathcal{Z}_t, \tag{8}$$

and the loadings for the IRP component of $y_{n,t}$ are obtained from Equations (3), (6), and (8):

$$y_{n,t}^{IRP} = y_{n,t} - y_{n,t}^{EC} \tag{9a}$$

$$= a_n - a_n^{EC} + (\boldsymbol{b}_n - \boldsymbol{b}_n{}^{EC})\mathcal{Z}_t \tag{9b}$$

$$= a_n^{IRP} + \boldsymbol{b}_n^{IRP}\mathcal{Z}_t. \tag{9c}$$

Hence, the IRP is the difference between the fitted inflation swaps and the expected inflation rate (EC). It will be positive (negative) when the inflation swaps are above (below) the expected inflation rate.

2.2 Dynamic System

2.2.1 Risk factors

Our set of risk factors includes (up to) ten variables, divided in four categories: (i) two global variables $\mathbf{w}_t = [g_t^w, p_t^o]'$, being global real economic activity (percent deviations from trend), and the log of real oil prices, respectively; (ii) three country-specific macro variables $\mathbf{m}_t = [\pi_t, g_t, i_t]'$, containing inflation, π_t, economic activity, g_t, and the policy rate, i_t; (iii) three ILS principal components,[4] the level, l_t, the slope, s_t, and the curvature, c_t, all stacked in $\mathbf{p}_t = [l_t, s_t, c_t]'$ and equal to $\mathbf{p}_t = W_p Y_t$, and (iv) two stochastic trends, one for inflation, π_t^*, and another for the equilibrium real interest rate (or natural rate of interest), r_t^*, collected in the vector $\boldsymbol{\xi}_t = [\pi_t^*, r_t^*]'$.

The general principle behind our choice of risk factors was to have a comparable set of variables across regions aligned with our research question. The motivation for selecting the two international variables builds on several recent studies investigating the interplay between oil prices

[4] Section 3.1.1 reports the loadings of the three principal components and shows that they can be interpreted as the level, slope, and curvature.

and inflation expectations in the US and the EA, even if we refrain from building a model with structural identification of the oil shocks. Two recent studies use a structural model to assess how oil (or gasoline) shocks affect US consumer inflation expectations (Aastveit, Bjørnland, and Cross 2023; Kilian and Zhou 2022). They report that oil shocks impact short-term inflation expectations. For the EA, Baumann et al. (2021) investigate the link between inflation expectations and oil prices. Using a structural identification for oil shocks, they conclude that only short-term expectations react to global real activity and oil-specific demand shocks.[5]

For the area-specific macroeconomic factors, we followed the macro-finance literature (see Ang and Piazzesi 2003) and selected variables that summarize the interaction between local demand, local inflation, and local monetary policy variables. Local macroeconomic variables (or news related to them) have been found to affect market-based inflation expectations. For example, Galati, Poelhekke, and Zhou (2011) show that ILS-based indicators of inflation expectations became more sensitive to news about inflation and other domestic macroeconomic variables. Finally, the two stochastic trends work as the long-term time-varying expected value of inflation, the ILS PCs (the stochastic trend for inflation), and the short-term interest rates (the stochastic trend for the real rate).

Including stochastic trends in the analysis offers several advantages. First, it allows for decomposing ILS rates into long-term and short-term determinants, which is central to our research question. Second, it helps avoid issues related to implausible behavior of the model's deterministic trend, which builds on the initial conditions of state dynamics.[6] In our context, much of the low-frequency behavior of ILS rates would be explained by initial conditions of the stationary factors. Allowing for time-varying intercepts for inflation and interest rates enables the data to determine how much of the low-frequency variation is linked to stochastic trends. In this regard, it is important to include information about the potential fluctuations of these stochastic trends to prevent them from capturing spurious variation and overfitting the model. Surveys, as discussed infra, play this role in our econometric framework. To further justify the choice of our set of factors, we conducted a model-free analysis of unspanned macro risk variation in both economic areas. The results indicate that, to various degrees, our factors have some explanatory power on ex-post excess returns movements, on top of the three spanned factors, see Section A.2 of the appendix.

The time series evolution[7] of the global factors, the country-specific variables, and the principal components are summarized via a vector error-correction setting:

$$\begin{bmatrix} \mathbf{w}_t \\ \mathbf{m}_t \\ \mathbf{p}_t \end{bmatrix} = \underbrace{\sum_{\upsilon=1}^{\Upsilon} \begin{bmatrix} \boldsymbol{\Phi}_\upsilon^{w} & 0 & 0 \\ \boldsymbol{\Phi}_\upsilon^{mw} & \boldsymbol{\Phi}_\upsilon^{m} & 0 \\ \boldsymbol{\Phi}_\upsilon^{pw} & \boldsymbol{\Phi}_\upsilon^{pm} & \boldsymbol{\Phi}_\upsilon^{p} \end{bmatrix} \begin{bmatrix} \mathbf{w}_{t-\upsilon} - \boldsymbol{\mu}^{w} - \boldsymbol{\Gamma}^{w}\boldsymbol{\xi}_t \\ \mathbf{m}_{t-\upsilon} - \boldsymbol{\mu}^{m} - \boldsymbol{\Gamma}^{m}\boldsymbol{\xi}_t \\ \mathbf{p}_{t-\upsilon} - \boldsymbol{\mu}^{p} - \boldsymbol{\Gamma}^{p}\boldsymbol{\xi}_t \end{bmatrix}}_{\text{Short-term component}} +$$

$$+ \underbrace{\begin{bmatrix} \boldsymbol{\mu}^{w} \\ \boldsymbol{\mu}^{m} \\ \boldsymbol{\mu}^{p} \end{bmatrix} + \begin{bmatrix} \boldsymbol{\Gamma}^{w} \\ \boldsymbol{\Gamma}^{m} \\ \boldsymbol{\Gamma}^{p} \end{bmatrix} \boldsymbol{\xi}_t}_{\text{Long-term component}} + \underbrace{\begin{bmatrix} \boldsymbol{\Sigma}^{w} & 0 & 0 \\ \boldsymbol{\Sigma}^{mw} & \boldsymbol{\Sigma}^{m} & 0 \\ \boldsymbol{\Sigma}^{pw} & \boldsymbol{\Sigma}^{pk} & \boldsymbol{\Sigma}^{p} \end{bmatrix} \begin{bmatrix} \boldsymbol{\varepsilon}_t^{w} \\ \boldsymbol{\varepsilon}_t^{m} \\ \boldsymbol{\varepsilon}_t^{p} \end{bmatrix}}_{\text{Innovations}}, \quad \begin{bmatrix} \boldsymbol{\varepsilon}_t^{w} \\ \boldsymbol{\varepsilon}_t^{m} \\ \boldsymbol{\varepsilon}_t^{p} \end{bmatrix} \sim \mathcal{N}(0, I), \tag{10}$$

[5] Among others, Galati et al. (2018) reach a similar conclusion about the relationship between oil prices and short-term inflation expectations, but without using any structural identification of the oil market's shocks.

[6] As highlighted by Giannone, Lenza, and Primiceri (2019), Sims (2000) demonstrates this point by simulating artificial data from a random walk process. He then estimates an AR(1) model on the simulated data and shows that the estimated model attributes a large fraction of the low-frequency behavior of the series to the deterministic component of the AR(1).

[7] Note that the probability measure of reference is the historical one, but, to ease notation, we drop the superscript $\mathbb{P}$.

where $\boldsymbol{I}$ is an identity matrix of conformable dimension. Equation (10) is the multivariate and multi-lag extension of Kozicki and Tinsley (2001) and Dewachter and Iania (2011).

2.2.2 Short-term dynamics

The short-term component of Equation (10) describes how innovations to the system—brought about by the three sets of normally distributed shocks $\boldsymbol{\varepsilon}_t^w$, $\boldsymbol{\varepsilon}_t^m$, and $\boldsymbol{\varepsilon}_t^p$—propagate through time, causing a deviation of $\mathbf{w}_t$, $\mathbf{m}_t$, and $\mathbf{p}_t$ from their long-run values. Under suitable stability conditions, see infra, if no additional shocks hit the system, these deviations converge to zero.

Three points are worth mentioning regarding the parameters included in the short-term and innovations components. First, Σ^w, Σ^m, and Σ^p are lower triangular, reflecting a Cholesky ordering of the shocks, with international factors being the most exogenous, followed by the country-specific factors and the principal components. For the international factors, the most exogenous innovation is real economic activity, while the ordering of country-specific factors follows the literature in assuming that supply shocks are the most exogenous, followed by demand shocks. Second, we impose that the ILS factors do not affect the evolution of the macro variables and that local macro variables do not affect the evolution of the international factors. This restriction is guaranteed by the Cholesky ordering and by imposing that $\Phi_v^{wm}, \Phi_v^{wp}, \Phi_v^{mp} = \mathbf{0}\ \forall v$.[8] Third, following Joslin, Le, and Singleton (2013), we select a parsimonious one-lag structure on the ILS principal components by imposing that $\Phi_v^p = \mathbf{0}\ \forall v > 1$.

2.2.3 Long-run dynamics

To analyze the long-run behavior of $\mathbf{w}_t$, $\mathbf{m}_t$, and $\mathbf{p}_t$, we specify a random walk process for the stochastic trends, and we conveniently write Equation (10) in more compact form by defining $\mathbf{x}_t = [\mathbf{w}_t', \mathbf{m}_t', \mathbf{p}_t']'$:

$$\mathbf{x}_t = \sum_{v=1}^{\Upsilon} \Phi_v^x (\mathbf{x}_{t-v} - \boldsymbol{\mu}^x - \Gamma^x \boldsymbol{\xi}_t) + \boldsymbol{\mu}^x + \boldsymbol{\Gamma}^x \boldsymbol{\xi}_t + \Sigma^x \boldsymbol{\varepsilon}_t^x, \quad \boldsymbol{\varepsilon}_t^x \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}) \tag{11a}$$

$$\boldsymbol{\xi}_t = \boldsymbol{\xi}_{t-1} + \Sigma^\xi \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}), \tag{11b}$$

where superscript x denotes the compact vectors/matrices of Equation (10). The long-run conditional expectation of $\mathbf{x}_{t+j}$, i.e. $\lim_{j\to\infty} \mathbb{E}^{\mathbb{P}}[\mathbf{x}_{t+j}|\mathcal{I}_t]$, relies on roots of the polynomial $\Phi^x(\mathcal{L}) = \Phi_1^x + \Phi_2^x \mathcal{L} + \ldots + \Phi_{\Upsilon-1}^x \mathcal{L}^{\Upsilon-1}$, defined in the lag operator $\mathcal{L}\mathbf{x}_t = \mathbf{x}_{t-1}$. If all roots of $I - \Phi^x(\mathcal{L})$ lie outside the unit circle, the short-term component of Equation (11a) (or equivalently Equation (10)) shrinks to zero. As a consequence:

$$\lim_{j\to\infty} \mathbb{E}[\mathbf{x}_{t+j}|\mathcal{I}_t] = \boldsymbol{\mu}^x + \Gamma^x \boldsymbol{\xi}_t, \tag{12}$$

which implies that the long-run expectations of the macro variables and the principal components incorporate a constant and a stochastic part, equal to $\Gamma^x \boldsymbol{\xi}_t$ and governed by the cointegration matrices $\Gamma^x = [\Gamma^{w'}, \Gamma^{m'}, \Gamma^{p'}]'$, see Equation (10), and the evolution of the stochastic trends $\boldsymbol{\xi}_t$, see Equation (11b). Our three sets of assumptions for the long-run components of x_t are summarized below:

$$\boldsymbol{\mu}^{\mathrm{w}} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \ \boldsymbol{\mu}^{\mathrm{m}} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \ \boldsymbol{\mu}^{\mathrm{p}} = \begin{bmatrix} \mu^l \\ \mu^s \\ \mu^c \end{bmatrix}, \tag{13a}$$

[8] The results are equivalent when we relax the condition that $\Phi_v^{wm} = \mathbf{0}$.

$$\Gamma^{\mathrm{w}} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \; \Gamma^{\mathrm{m}} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}, \; \Gamma^{\mathrm{p}} = \begin{bmatrix} \gamma^{l} & 0 \\ \gamma^{s} & 0 \\ \gamma^{c} & 0 \end{bmatrix}. \tag{13b}$$

First, we assume stationarity for the demeaned global economic factors, $\Gamma^{\mathrm{w}} = \mathbf{0}$ and $\boldsymbol{\mu}^{\mathrm{w}} = \mathbf{0}$. Second, for the local economic factors, we assume that the Fisher (1896) equation holds in the long run, and we demeaned the measure of real activity, $\boldsymbol{\mu}^{\mathrm{m}} = \mathbf{0}$, implying:

$$\lim_{j\to\infty} \mathbb{E}[\pi_{t+j}|\mathcal{I}_t] = \pi_t^* \tag{14a}$$

$$\lim_{j\to\infty} \mathbb{E}[g_{t+j}|\mathcal{I}_t] = 0 \tag{14b}$$

$$\lim_{j\to\infty} \mathbb{E}[i_{t+j}|\mathcal{I}_t] = \pi_t^* + r_t^*. \tag{14c}$$

Finally, to derive the long-term relationships for the ILS factors, we start from the intuition that, under the historical probability measure, for any maturity, the long-run expected value of the ILS rates converges to

$$\lim_{j\to\infty} \mathbb{E}[y_{n,t+j}|\mathcal{I}_t] = \psi_n + \pi_t^*, \tag{15}$$

where ψ_n is a maturity-dependent constant accounting for potential premiums linked, for example, to liquidity. Next, since l_t, s_t, and c_t are a linear combination of the N ILS rates, collected in $Y_t = [y_{1,t}, \ldots, y_{N,t}]'$, we obtain:

$$\lim_{j\to\infty} \mathbb{E}[l_{t+j}|\mathcal{I}_t] = \lim_{j\to\infty} \mathbb{E}[w_l Y_{t+j}|\mathcal{I}_t] = w_l(\boldsymbol{\psi} + \boldsymbol{\iota}\pi_t^*) = w_l\boldsymbol{\psi} + w_l\boldsymbol{\iota}\pi_t^* \tag{16a}$$

$$\lim_{j\to\infty} \mathbb{E}[s_{t+j}|\mathcal{I}_t] = \lim_{j\to\infty} \mathbb{E}[w_s Y_{t+j}|\mathcal{I}_t] = w_s(\boldsymbol{\psi} + \boldsymbol{\iota}\pi_t^*) = w_s\boldsymbol{\psi} + w_s\boldsymbol{\iota}\pi_t^* \tag{16b}$$

$$\lim_{j\to\infty} \mathbb{E}[c_{t+j}|\mathcal{I}_t] = \lim_{j\to\infty} \mathbb{E}[w_c Y_{t+j}|\mathcal{I}_t] = w_c(\boldsymbol{\psi} + \boldsymbol{\iota}\pi_t^*) = w_c\boldsymbol{\psi} + w_c\boldsymbol{\iota}\pi_t^*, \tag{16c}$$

where $\boldsymbol{\iota}$ is an N-dimensional column vector of ones and $\boldsymbol{\psi}$ is defined as $\boldsymbol{\psi} = [\psi_1, \psi_2, \ldots, \psi_N]'$. The long-run relationships expressed in Equations (16a)–(16c) have the following implication for $\boldsymbol{\mu}^{\mathrm{p}}$ and Γ^{p}. First, the elements of Γ^{p} governing the cointegration relationship between the stochastic trend of inflation and the ILS principal components (γ^l, γ^s, γ^c) are fixed to $w_l\boldsymbol{\iota}$, $w_s\boldsymbol{\iota}$, and $w_c\boldsymbol{\iota}$, respectively. Second, the constant parts of the long-term components of $\mathbf{p}_t$, i.e., the elements of $\boldsymbol{\mu}^{\mathrm{p}}$, are simply a linear combination of the individual ψ's, which we treat as unknown parameters.

By defining $\mathbf{x}_t^{1:\mathrm{Y}} = [\mathbf{x}_t', \ldots, \mathbf{x}_{t-\mathrm{Y}+1}']'$ and $\mathcal{Z}_t = [\mathbf{x}_t^{1:\mathrm{Y}'}, \boldsymbol{\xi}_t']'$, and stacking the shocks in $\boldsymbol{\varepsilon}_t^{\mathbb{P}} = [\boldsymbol{\varepsilon}_t^{x'}, \boldsymbol{\eta}_t']'$, we express the dynamic system in its companion form, which corresponds to Equation (4b) under $\mathbb{P}$:

$$\mathcal{Z}_t = \boldsymbol{\mu}^{\mathbb{P}} + \Phi^{\mathbb{P}}\mathcal{Z}_{t-1} + \Sigma\boldsymbol{\varepsilon}_t^{\mathbb{P}}, \quad \boldsymbol{\varepsilon}_t^{\mathbb{P}} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}). \tag{17}$$

The next section lays out the identification restrictions to Equation (4b) under $\mathbb{Q}$.

2.3 Canonical Representation

We adopt the normalization of Joslin, Singleton, and Zhu (2011), henceforth JSZ, and extend the setting of Bauer and Rudebusch (2020) to two stochastic trends, which we apply to the ILS market. The JSZ identification technique dramatically reduces the number of parameters of Equation (4b) under $\mathbb{Q}$ via two sets of restrictions. First, it imposes the so-called spanning conditions, that is, under $\mathbb{Q}$ only a few linear combinations of Y_t ($\mathbf{p}_t$ for our model) span the cross-section of ILS rates over time. Hence,

$$\mathbb{E}^{\mathbb{Q}}[y_{n,t+j}|\mathcal{Z}_t] = \mathbb{E}^{\mathbb{Q}}[y_{n,t+j}|\mathbf{p}_t]. \tag{18}$$

To satisfy condition (18), we impose that, in the risk-neutral dynamics, Equations (4a) and (4b) depend only on $\mathbf{p}_t$:

$$y_{1,t} = \rho_0 + \boldsymbol{\rho}_1^{p'} \mathbf{p}_t \tag{19a}$$

$$\mathbf{p}_t = \boldsymbol{\mu}^{\mathbb{Q},p} + \Phi^{\mathbb{Q},p} \mathbf{p}_{t-1} + \tilde{\Sigma}^p \boldsymbol{\varepsilon}_t^{\mathbb{Q}}, \quad \boldsymbol{\varepsilon}_t^{\mathbb{Q}} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}), \tag{19b}$$

where (i) the 3×1 vectors $\boldsymbol{\rho}_1^{p'}$ and $\boldsymbol{\mu}^{\mathbb{Q},p}$ and the 3×3 matrices $\Phi^{\mathbb{Q},p}$ are the non-zero components of $\boldsymbol{\rho}_1'$, $\boldsymbol{\mu}^{\mathbb{Q}}$, and $\Phi^{\mathbb{Q}}$ in Equation (4b) corresponding to $\mathbf{p}_t$, and (ii) the 3×10 matrix $\tilde{\Sigma}^p$ corresponds to the non-zero elements of the last three rows of Σ, see Equation (10). Equations (19a) and (19b) imply that the ILS rates are an affine function of $\mathbf{p}_t$. Hence, Equation (6) simplifies to:

$$y_{n,t} = a_n^p + \boldsymbol{b}_n^p \mathbf{p}_t, \tag{20}$$

where the loadings are obtained similarly as in Equations (7a) and (7b), namely, $a_n^p = -n^{-1} a_n^p$ and $\boldsymbol{b}_n^p = -n^{-1} b_n^p$:

$$a_n^p = a_{n-1}^p + b_{n-1}^p \boldsymbol{\mu}^{\mathbb{Q},p} + \frac{1}{2} b_{n-1}^p \Omega^{\frac{1}{2}} (b_{n-1}^p \Omega^{\frac{1}{2}})' - \rho_0 \tag{21a}$$

$$b_n^p = (\Phi^{\mathbb{Q},p})' b_{n-1}^p - \boldsymbol{\rho}_1^p. \tag{21b}$$

with $\Omega = \tilde{\Sigma}^p \tilde{\Sigma}^{p'}$. Looking at Equations (19a), (19b), and (20), we notice that the spanning condition (18) is trivially satisfied as ILS rates at any maturity depend only on $\mathbf{p}_t$, which in turn, under $\mathbb{Q}$ depends only on its lagged value. The spanning assumption imposes a set of zero restrictions on the stochastic discount factor and the prices of risks, see Equations (5a) and (5b), implying that:

$$M_{t,t+1} = \exp\left(-y_{1,t} - \frac{1}{2} \boldsymbol{\Lambda}_t^{p'} \boldsymbol{\Lambda}_t^p - \boldsymbol{\Lambda}_t^{p'} \boldsymbol{\varepsilon}_t\right) \tag{22a}$$

$$\boldsymbol{\Lambda}_t^p = \Omega^{-\frac{1}{2}} (\boldsymbol{\lambda}_0^p + \boldsymbol{\lambda}_1^p \mathcal{Z}_t) \tag{22b}$$

$$\boldsymbol{\lambda}_0^p = \boldsymbol{D}_p \boldsymbol{\mu}^{\mathbb{P}} - \boldsymbol{\mu}^{\mathbb{Q},p} \tag{22c}$$

$$\boldsymbol{\lambda}_1^p = \boldsymbol{D}_p \Phi^{\mathbb{P}} - [\mathcal{O}\, \Phi^{\mathbb{Q},p}], \tag{22d}$$

with $\boldsymbol{\varepsilon}_t = \Omega^{-\frac{1}{2}} \tilde{\boldsymbol{\varepsilon}}_t^{\mathbb{Q}}$, $\tilde{\boldsymbol{\varepsilon}}_t^{\mathbb{Q}} = \tilde{\Sigma}^p \boldsymbol{\varepsilon}_t^{\mathbb{Q}}$, $\mathcal{O}$ a zero-entry matrix of conformable dimension, and $\boldsymbol{D}_p = [\mathcal{O}\, \boldsymbol{I}_{3\times 3}]$ a matrix selecting the last three rows of $\boldsymbol{\mu}^{\mathbb{P}}$ and $\Phi^{\mathbb{P}}$, i.e. the parts related to $\mathbf{p}_t$. In a nutshell, as the cross-section of ILS rates is summarized by $\mathbf{p}_t$, only shocks to those principal components are priced, see Equation (22a). However, via Equations (22a) and (22b), unspanned factors play a role in the pricing of those shocks, which are weighted by the 3×1 prices of risk vector $\mathbf{\Lambda}_t^p$, that in turn is linked to all the variables of the dynamic system (17) via the matrix $\boldsymbol{\lambda}_1^p$ of Equation (22b).

To identify the risk-neutral parameters, we follow JSZ and impose a second set of restrictions identifying restrictions on the $\mathbb{Q}$-dynamics, so that the real-world dynamics can be estimated without restrictions. We start with $N=3$ latent factors χ_t and impose the following restrictions on their $\mathbb{Q}$-dynamics:

$$y_{1,t} = \iota\chi_t,\ \boldsymbol{\mu}^\chi = \begin{bmatrix} \boldsymbol{\kappa}^\chi \\ 0 \\ 0 \end{bmatrix},\ \mathbf{\Phi}^\chi = \begin{bmatrix} \lambda_1^\chi & 0 & 0 \\ 0 & \lambda_2^\chi & 0 \\ 0 & 0 & \lambda_3^\chi \end{bmatrix}, \tag{23}$$

where ι is 1×3 vector of ones and the $\boldsymbol{\lambda}^\chi$s are assumed to be less than one in absolute value. Subsequently, we obtain the term structure loadings related to χ_t, A_χ, and B_χ, and use the definition of $\mathbf{p}_t = W_p Y_t$ to obtain the loadings for $\mathbf{p}_t$ in function of the parameters in (23):

$$Y_t = (I - B_\chi(W_p B_\chi)^{-1} W_p)A_\chi + B_\chi(W_p B_\chi)^{-1}\mathbf{p}_t, \tag{24}$$

where A_χ and B_χ are the affine yield loadings on χ_t and $W_p = [w_l', w_s', w_c']'$.

3 Empirical Implementation

3.1 Data and Preliminary Statistics

We estimate the model separately for the EA and the US using a mix of quarterly and monthly data from January 2005 to August 2023. Our empirical analysis combines four types of information: data on ILS rates, macroeconomic variables, survey forecasts, and model-based long-run trend estimates from the literature.

3.1.1 ILS data

The swap series are from Bloomberg. We construct a monthly series by taking the end-of-the-month data from the daily ILS rates with maturity 1 to 10, 12, 15, 20, 25, and 30 years, collected in the vector Y_t. Table 1A and B report the summary statistics of the EA (US) 1-, 5-, 10-, and 30-year maturity ILS rates. In both regions, ILS rates are characterized by upward-sloping averages, decreasing volatility, short-term leptokurtosis (fat tails), and increasing persistence (EA 30-year rates are not more persistent than 10-year inflation swaps). US ILS rates are, on average, higher and less persistent than the EA counterparts. Short-term (long-term) US ILS rates[9] are 0.1 pp. (0.4 pp.) higher than the European ones: 1.8% versus 1.7% (2.5% versus 2.1%). Average US ILS rates are above the 2% target of the Federal Reserve. This difference might be due to the Federal Reserve's inflation target being based on the Personal Consumption Expenditure index, which delivers, on average, lower inflation rates than the Consumer Price Index (CPI), the underlying index in ILS contracts. Panels A and B of Figure 1 indicate that EA and US long-term ILS rates, i.e., above ten years of maturity, display a declining trend from around 2013 until September 2020. The recent COVID crisis and the Russian aggression against Ukraine reverted this trend, causing the longest observed inversion in the term structure of ILS rates in both regions.

From vector Y_t, we extract for each country the first three principal components $PC1_t$, $PC2_t$, and $PC3_t$, whose statistics, plots and loadings are reported in Table 1C and D, and

[9] Short-term and long-term refer to 1-year and 30-year maturity, respectively.

Table 1. Summary statistics: ILS rates and PCs

	(A) EA ILS						(B) US ILS					
	μ	σ	s	k	ρ_1	ρ_3	μ	σ	s	k	ρ_1	ρ_3
1-year	0.017	0.013	1.961	8.902	0.945	0.839	0.018	0.012	−1.087	8.246	0.908	0.626
5-year	0.017	0.006	0.018	2.499	0.960	0.879	0.022	0.005	−0.661	4.661	0.919	0.665
10-year	0.018	0.005	−0.282	2.128	0.970	0.905	0.024	0.004	−0.640	2.992	0.923	0.751
30-year	0.021	0.004	−0.540	2.428	0.966	0.885	0.025	0.004	−0.241	2.300	0.936	0.822

	(C) EA PCs							(D) US PCs						
	μ	σ	s	k	ρ_1	ρ_3	$\frac{\lambda_j}{\sum \lambda_n}$	μ	σ	s	k	ρ_1	ρ_3	$\frac{\lambda_j}{\sum \lambda_n}$
PC 1	0.063	0.024	0.493	3.530	0.960	0.871	0.898	0.075	0.021	−0.827	5.827	0.919	0.663	0.856
PC 2	−0.028	0.008	0.899	4.719	0.948	0.864	0.094	−0.048	0.008	0.132	2.138	0.930	0.858	0.132
PC 3	0.011	0.002	0.358	2.351	0.844	0.728	0.006	−0.002	0.002	−0.566	6.618	0.638	0.259	0.007

Notes: Panels (A) and (B) report the summary statistics of the EA and US ILS rates with maturities of 1, 5, 10, and 30 years. Panels (C) and (D) report the summary statistics of the EA and US first three principal components (PCs) of all the ILS rates. Symbols: μ = average; σ = standard deviation; s = skewness; k = kurtosis; ρ_1 = first order autocorrelation; ρ_3 = third order autocorrelation; λ_j = eigenvalue associated with the j-*th* principal component.

in panels C to F of Figure 1, respectively. Notably, in both regions, the first three principal components capture more than 99% of the total variation, with the first factor responsible for about 90% of the variation in the EA and 86% for the US.

For both ILS term structures, the loadings of the first factor are the (exponentially decaying) weighted averages of all ILS rates, while the second factors load negatively on short-term maturities and positively on long-term ones, giving them the interpretation of level and slope factors. The third factor is the difference between (i) the average of long- and short-term ILS rates and (ii) medium-term ones, i.e., the common shape of yield curve curvature factors. By summing up the weights for each factor/country, that is, w_l, w_s, and w_c, the resulting entries for the cointegration matrices Γ^{p}s of Equation (13b) are:

$$\Gamma^{\mathrm{p},EA} = \begin{bmatrix} 3.8 & 0 \\ 0.8 & 0 \\ 0.3 & 0 \end{bmatrix}, \quad \Gamma^{\mathrm{p},US} = \begin{bmatrix} 3.5 & 0 \\ 1.6 & 0 \\ -0.1 & 0 \end{bmatrix}.$$

3.1.2 Macroeconomic data

Macroeconomic variables are divided into international and country-specific factors. The international variables are (i) the Kilian (2009) index of global real economic activity (percent deviation from trend) and (ii) the monthly log of real oil price (spot), that is, the West Texas Intermediate for the US and the Brent for the EA, both deflated by the respective local CPIs.[10]

Country-specific variables are obtained from the Federal Reserve Economic Data (FRED) database maintained by the Federal Reserve Bank of St. Louis for the US and from the ECB Data Portal for the EA.[11] We proxy the inflation factor via the month-on-month growth rate of the inflation index underlying the ILS contracts, that is, the seasonally-adjusted Harmonized Index of Consumer Prices excluding tobacco for the EA and the CPI for the US, while the month-on-month growth rate of the Industrial Production Index gives

10 The results are equivalent when using nominal oil prices instead.
11 The data codes are listed in Section A.1.

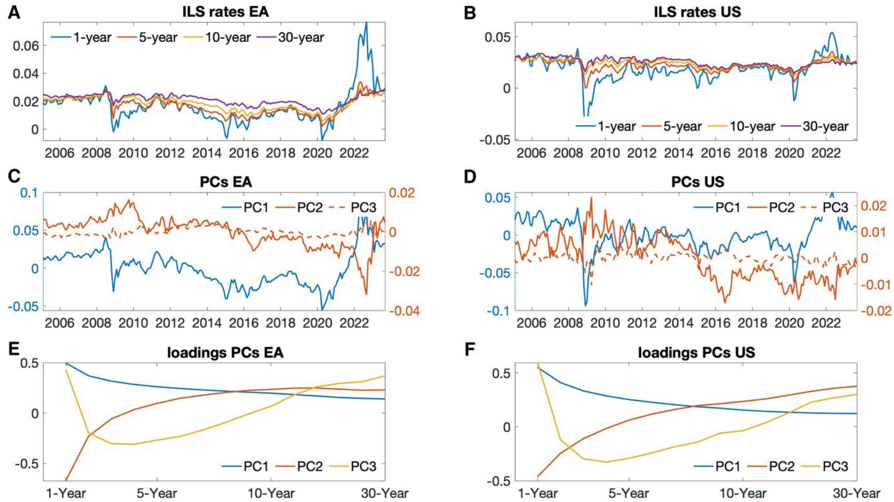

Figure 1. ILS rates, principal components, and loadings.

Notes: Panels (A) and (B) plot the EA and US ILS rates with maturities of 1, 5, 10, and 30 years. Panels (C) and (D) depict the first three principal components of all the EA and US ILS rates, while Panels (E) and (F) report their respective loadings.

the real activity measure.[12] As risk-free rates, we use the one-year T-bill rate for the US and the one-year OIS rate for the EA.

3.1.3 Surveys and model-based series

We exploit the information contained in surveys and the quarterly natural rate of interest constructed by Holston, Laubach, and Williams (2017) to (i) identify the two stochastic trends and (ii) identify the value of real activity at the beginning of the COVID-19 period.

To fix notations, the surveys forecasts for a variable x are labeled as $\mathbb{E}_t^{\mathcal{S}} x_{t+t_1 \to t+t_2}$, where (i) $\mathcal{S}$ indicates the source from where the subjective forecasts are obtained, and (ii) $t+t_1$ and $t+t_2$ are the beginning and the end of the forecasting period, respectively, expressed in months.

For each country, we use two survey providers: the Survey of Professional Forecasters (SPF) and the Consensus Economics (CE) forecasts, that is, $\mathcal{S} = \{SPF, CE\}$. SPF data are provided by the European Central Bank (ECB) for the EA and by the Philadelphia Federal Reserve for the US. From the ECB (Philadelphia FED) SPFs, we select the year-on-year inflation forecast in two and five years (average inflation over the next five and ten years), which we label $\mathbb{E}_t^{SPF} \pi_{t+t_1 \to t+t_2}$, with $t_1 = \{12, 48\}$ and $t_2 = \{24, 60\}$ ($t_1 = \{1, 1\}$ and $t_2 = \{60, 120\}$) months.

We use two sets of CE surveys, the short-term and long-term forecasts. Short-term CE forecasts for the next-year inflation rate, $\mathbb{E}_t^{CE} \pi_{t+1 \to t+12}$, industrial production, $\mathbb{E}_t^{CE} g_{t+1 \to t+12}$, and one-year interest rates, $\mathbb{E}_t^{CE} i_{t+1 \to t+12}$, are proxied by an extrapolation from (i) the current and next calendar year forecasts and (ii) from the next-year forecast for the 3-month and 10-year interest rates, respectively:

[12] We find similar results with the following local EA inflation measures: (i) the HICP excluding energy and food (ICP.M.U2.Y.XEF000.3.INX) and (ii) the HICP—All-items.

$$\mathbb{E}_t^{CE}\pi_{t+1\to t+12} = \frac{(13-\mathsf{m}(t))-1}{12}\mathbb{E}_t^{CE}\pi_{\mathsf{y}(t)} + \frac{(\mathsf{m}(t)-1)}{12}\mathbb{E}_t^{CE}\pi_{\mathsf{y}(t)+1} \tag{25a}$$

$$\mathbb{E}_t^{CE}g_{t+1\to t+12} = \frac{(13-\mathsf{m}(t)-1)}{12}\mathbb{E}_t^{CE}g_{\mathsf{y}(t)} + \frac{(\mathsf{m}(t)-1)}{12}\mathbb{E}_t^{CE}g_{\mathsf{y}(t)+1} \tag{25b}$$

$$\mathbb{E}_t^{CE}i_{t+12} = \mathbb{E}^{CE}i_{t+12}^3 + \frac{\mathbb{E}_t^{CE}i_{t+12}^{120} - \mathbb{E}_t^{CE}i_{t+12}^3}{120-3}(12-3). \tag{25c}$$

In Equations (25a) and (25b), $\mathsf{y}(t)$ and $\mathsf{m}(t)$ are the calendar years and months related to time t, respectively. The expression approximates fixed-horizon forecasts as a weighted average of fixed-event forecasts (see Dovern, Fritsche, and Slacalek 2012). Equation (25c) approximates the forecast of the 1-year interest rate, $\mathbb{E}_t^{CE}i_{t+12}$, as the weighted average of the forecast of the 3-month and 10-year interest rates in one year, $\mathbb{E}_t^{CE}i_{t+12}^3$ and $\mathbb{E}_t^{CE}i_{t+12}^{120}$, respectively. The implicit assumption is that the forecasters-implied yield curve has the same slope for all maturities. Long-term CE forecasts are expectations in five to ten years of short-term interest rates and inflation, $\mathbb{E}_t^{CE}i_{t+60\to t+120}^3$ and $\mathbb{E}_t^{CE}\pi_{t+60\to t+120}$.

Finally, our last observed variable is the quarterly natural interest rate constructed by Holston, Laubach, and Williams (2017), $\tilde{r}_t^*$, which helps identify the stochastic trend for the interest rate.

3.2 Estimation Strategy

The estimation of our term structure model involves two steps: (i) the identification of the stochastic trends and the fitting of risk factors' dynamic system (17) and (ii) the pricing of the ILS contracts via the multivariate version of Equation (20). The first step is performed in a state-space setting, while the second maximizes the fit of ILS rates via a (constrained) linear system. We estimate the system outlined in Section 2.2 via a state-space setting, whereby the state variables, $\mathcal{Z}_t$, see (17), are linearly related to a set of observable variables, $\mathcal{Y}_t$:

$$\mathcal{Y}_t = \mathbf{a} + \mathbf{B}\mathcal{Z}_t + \Sigma^{\mathcal{Y}}\boldsymbol{\varepsilon}_t^{\mathcal{Y}}, \quad \boldsymbol{\varepsilon}_t^{\mathcal{Y}} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}), \tag{26}$$

where $\Sigma^{\mathcal{Y}}$ is a diagonal matrix of standard deviations. The observed series and their corresponding loadings, $\mathbf{a}$ and $\mathbf{B}$, are divided in four blocks:

$$\mathcal{Y}_t' = [\,\mathcal{Y}_t^{\mathcal{O}'} \quad \mathcal{Y}_t^{\pi'} \quad \mathcal{Y}_t^{g'} \quad \mathcal{Y}_t^{i'}\,] \tag{27a}$$

$$\mathbf{a}' = [\,\mathbf{a}^{\mathcal{O}'} \quad \mathbf{a}^{\pi'} \quad \mathbf{a}^{g'} \quad \mathbf{a}^{i'}\,] \tag{27b}$$

$$\mathbf{B}' = [\,\mathbf{B}^{\mathcal{O}'} \quad \mathbf{B}^{\pi'} \quad \mathbf{B}^{g'} \quad \mathbf{B}^{i'}\,]. \tag{27c}$$

The first block, $\mathcal{Y}_t^{\mathcal{O}}$, includes the observed data, i.e. $\mathbf{w}_t$, $\mathbf{m}_t$, and $\mathbf{p}_t$:

$$\mathcal{Y}_t^{\mathcal{O}} = [\,p_t^o \quad g_t^w \quad \pi_t \quad g_t \quad i_t \quad l_t \quad s_t \quad c_t\,]' \tag{28a}$$

$$\mathbf{a}^{\mathcal{O}} = [0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0]' \tag{28b}$$

$$\mathbf{B}^{\mathcal{O}} = [\,\iota^{\Delta p^{o'}} \quad \iota^{g^{w'}} \quad \iota^{\pi'} \quad \iota^{g'} \quad \iota^{i'} \quad \iota^{l'} \quad \iota^{s'} \quad \iota^{s'}\,]'. \tag{28c}$$

Since the stationary state variables are observed without error, the elements of $\mathbf{a}^{\mathcal{O}}$ are set to zero while $\mathbf{B}^{\mathcal{O}}$ is made of row vectors selecting the respective observed variables from the state vector, $\mathcal{Z}_t$. For example, for the macro series, these restrictions correspond to:

$$\iota^{\pi} = [0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad \ldots \quad 0] \tag{29a}$$

$$\iota^{g} = [0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad \ldots \quad 0] \tag{29b}$$

$$\iota^{i} = [0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad \ldots \quad 0]. \tag{29c}$$

To account for the large deviation of the industrial production series at the beginning of the COVID-19 period, we impose that g_t is not observed from March 2020 to August 2020. The third block, see infra, identifies g_t over that period. The second block includes the four inflation survey data from the SPF and from the CE:

$$\mathcal{Y}_t^{\pi} = [\mathbb{E}_t^{SPF}\pi_{t+t_1 \to t+t_2} \quad \mathbb{E}_t^{SPF}\pi_{t+t_3 \to t+t_4} \quad \mathbb{E}_t^{CE}\pi_{t+1 \to t+12} \quad \mathbb{E}_t^{CE}\pi_{t+61 \to t+120}]' \tag{30a}$$

$$\mathbf{a}^{\pi} = [\iota^{\pi}\bar{\mathbf{a}}_{t_1 \to t_2} \quad \iota^{\pi}\bar{\mathbf{a}}_{t_3 \to t_4} \quad \iota^{\pi}\bar{\mathbf{a}}_{1 \to 12} \quad \iota^{\pi}\bar{\mathbf{a}}_{61 \to 120}]' \tag{30b}$$

$$\mathbf{B}^{\pi} = [\iota^{\pi}\bar{\mathbf{B}}'_{t_1 \to t_2} \quad \iota^{\pi}\bar{\mathbf{B}}'_{t_3 \to t_4} \quad \iota^{\pi}\bar{\mathbf{B}}'_{1 \to 12} \quad \iota^{\pi}\bar{\mathbf{B}}'_{61 \to 120}]', \tag{30c}$$

where (i) for the EA (US) t_1, t_2, t_3, and t_4 are equal to 12, 24, 48, and 60 months (1, 60, 1, and 120 months), respectively, and ι^{π}, defined in Equation (29a) selects the inflation components of $\bar{\mathbf{a}}'_{\tau_1 \to \tau_2}$ and $\bar{\mathbf{B}}'_{\tau_1 \to \tau_2}$. For a generic forecasting horizon ranging from τ_1 to τ_2, the loadings $\bar{\mathbf{a}}'_{\tau_1 \to \tau_2}$ and $\bar{\mathbf{B}}'_{\tau_1 \to \tau_2}$ are the model-implied average forecasts of the state variables over the period $[\tau_1, \tau_2]$:

$$\bar{\mathbf{a}}_{\tau_1 \to \tau_2} = \frac{1}{\tau_2 - \tau_1 + 1} \sum_{H=\tau_1}^{\tau_2} \left[\sum_{h=1}^{H} \Phi^{h-1} \right] \boldsymbol{\mu} \tag{31a}$$

$$\bar{\mathbf{B}}_{\tau_1 \to \tau_2} = \frac{1}{\tau_2 - \tau_1 + 1} \sum_{H=\tau_1}^{\tau_2} \Phi^{H}. \tag{31b}$$

As we observed that the 12-month CE forecasts were substantially more volatile than the SPF's, we (i) scale its model-implied counterpart by a factor β^{CE} and (ii) we add to it a level factor, α^{CE}, to account for shifts in the level of the model-implied forecast brought about by the scaling factor β^{CE}.

The third block includes information related to industrial production around the first period of the COVID-19 crisis, from March 2020 to August 2020. During this period, we treat the series of Industrial Production Indexes as a latent factor which is noisily observed via (i) the realized month-on-month growth rate of the Industrial Production Index from March 2020 to August 2021 and (ii) its year-on-year CE forecast, $\mathbb{E}_t^{CE} g_{t+12}$. The resulting measured variables and loadings are:

$$\mathcal{Y}_t^{g} = [\tilde{g}_t \quad \mathbb{E}_t^{CE}\tilde{g}_{t+1 \to t+12}]' \tag{32a}$$

$$\mathbf{a}^{g} = [\alpha^{\tilde{g}} \quad \iota^{g}\bar{\mathbf{a}}_{1 \to 12}]' \tag{32b}$$

$$\mathbf{B}^{g} = [\beta^{\tilde{g}}\iota^{g} \quad \iota^{g}\bar{\mathbf{B}}'_{1 \to 12}]', \tag{32c}$$

where the parameters $\alpha^{\tilde{g}}$ and $\beta^{\tilde{g}}$ allow the observed industrial production growth to differ from the latent one, the loadings $\bar{\mathbf{a}}_{1 \to 12}$ and $\bar{\mathbf{B}}_{1 \to 12}$ are from Equations (31a) and (31b), and ι^{g} is defined in Equation (29b).

The last block, $\mathcal{Y}_t^{i}$, includes three types of information, coming from two sources: (i) the short- and long-term CE interest rates forecasts, and (ii) the Holston, Laubach, and

Williams (2017) measure of the equilibrium real rate. The resulting block of the measurement equations includes:

$$\mathcal{Y}_t^i = [\,\mathbb{E}_t^{CE} i_{t+12}^{12} \quad \mathbb{E}_t^{CE} i_{t+61\to t+120}^{12} \quad \tilde{r}_t^*\,]' \tag{33a}$$

$$\mathbf{a}^i = [\,\iota^i \bar{\mathbf{a}}_{12} \quad \iota^i \bar{\mathbf{a}}_{61\to 120} \quad \alpha^{\tilde{r}}\,]' \tag{33b}$$

$$\mathbf{B}^i = [\,\iota^i \bar{\mathbf{B}}'_{12} \quad \iota^i \bar{\mathbf{B}}'_{61\to 120} \quad \beta^{\tilde{r}}\,]', \tag{33c}$$

where the loading of the interest forecasts are obtained via the Equations (31a) and (31b), while $\alpha^{\tilde{r}}$ and $\beta^{\tilde{r}}$ account for potential differences in mean and volatility between our model-implied stochastic trend for the real rate and the Holston, Laubach, and Williams (2017)'s measure. Finally, given the identifying assumption related to the measurement equation, the matrix with the standard deviations of its errors is defined as:

$$\Sigma^{\mathcal{Y}} = \mathrm{diag}[\,0 \quad \sigma^g \quad \sigma^{SPF_1^\pi} \quad \dots \quad \sigma^{\tilde{r}}\,],$$

with all non-zero elements except for the observed inflation and short-term interest rate equations.

3.2.1 Likelihood function

We estimate the model composed by Equations (26) and (17) using a penalized likelihood approach, a method that is widely used in high-dimensional statistical modeling (see Tibshirani 1996; Fan and Li 2001, among others). The penalized log-likelihood is composed of two parts:

$$l \propto \underbrace{-\frac{1}{2}\sum_{t=\mathrm{Y}}^{T}(\ln|\boldsymbol{F}_t| + \boldsymbol{v}_t'\boldsymbol{F}_t^{-1}\boldsymbol{v}_t)}_{\text{KF-EPD}} + \underbrace{p(\Phi) + p(\Sigma^{\xi}) + p(\Sigma^{\mathcal{Y}}) + p(\alpha) + p(\beta)}_{\text{Penalty}}. \tag{34}$$

The component labeled KF-EPD is obtained by exploiting the Gaussian nature of the linear state-space model. It represents the sum of the likelihood of the Kalman Filter's prediction errors, $\boldsymbol{v}_t$, whose variance is given by $\boldsymbol{F}_t$. The functional forms of $\boldsymbol{v}_t$ and $\boldsymbol{F}_t$ derive from the Kalman Filter's recursions (Harvey 1990, 125).

The second part of the log-likelihood includes the penalties for the system's (block of) parameters. The first set of penalties relates to the elements of Φ. For these parameters, we adopt a $||\mathrm{L}||^2$ regularization, whereby the (scaled) elements of Φ are shrunk to zero via the following quadratic penalty function:

$$p(\Phi) = \sum_{v=1}^{\mathrm{Y}}\sum_{j=1}^{3}\sum_{i=1}^{3}\lambda_v(i,j)[v\frac{\tilde{\phi}_v(i,j)}{\varphi_{i,j}}]^2, \quad \lambda_v(i,j) = \begin{cases} \zeta_1^2 & \forall i = j \\ \zeta_1^2\zeta_2^2 & \text{otherwise.} \end{cases} \tag{35}$$

In Equation (35), the interpretation and values of the scaling coefficients $\varphi_{i,j}$, ζ_1, and ζ_2 are as follows. The coefficients $\varphi_{i,j}$ account for the fact that the macro series are not standardized and might thus have different sample variances. We set their values to the outer product of the standard deviations of $\hat{\mathbf{m}}_t$, the deviation of the macro variables from the stochastic trends extrapolated from the surveys series. To penalize model complexity, we divided $\varphi_{i,j}$ by the lag of the coefficients it refers to, implying stronger regularization for additional lags added to the model. The coefficients ζ_1 and ζ_2 control the overall shrinkage and the penalization for the interaction coefficients, respectively. Following the Bayesian literature, we set their values to $\zeta_1 = \zeta_2 = \sqrt{2}$.

The additional penalties relate to the identification of the stochastic trends. They are linked to (i) the standard deviations of the stochastic trends, Σ^{ξ}, and of the measurement Equation (26), $\Sigma^{\mathcal{Y}}$, and (ii) to the additional coefficients of the measurement equations, namely α^{CE}, β^{CE}, $\alpha^{\tilde{g}}$, $\beta^{\tilde{g}}$, $\alpha^{\tilde{g}}$, $\beta^{\tilde{g}}$, $\alpha^{\tilde{r}}$, and $\beta^{\tilde{r}}$. For the standard deviations, we use the following penalty function:

$$p(\boldsymbol{\sigma}^k) = \lambda^k[\ln \boldsymbol{\sigma}^k - \ln \bar{\boldsymbol{\sigma}}^k]^2, \ \ \{\lambda^k, \ln \bar{\boldsymbol{\sigma}}^k\} = \begin{cases} \{100, \ln \hat{\boldsymbol{\sigma}}^k\} & \forall k = \pi^*, r^* \\ \{1, -5\} & \text{otherwise}, \end{cases} \tag{36}$$

where $\bar{\boldsymbol{\sigma}}^{\pi^*}$ and $\bar{\boldsymbol{\sigma}}^{r^*}$ are extrapolated from the 2-year SPF and Holston, Laubach, and Williams's (2017) measure, respectively. Intuitively, the penalties on Equation (36) shrink the estimated parameters toward $\ln \hat{\boldsymbol{\sigma}}^k$, whereby a deviation from these centered values are penalized by a factor λ^k. Our penalty functions assign a great deal of importance to the survey information to the Holston, Laubach, and Williams' (2017) measure of the real rate, as it is highlighted by the values of $\ln \hat{\boldsymbol{\sigma}}^k$, which are relatively small or equal to their extrapolated values, and by the high values assigned to λ^k. For the additional coefficients of the measurement equations, we adopt the following penalty:

$$p(\boldsymbol{\theta}^k) = \lambda^k[\boldsymbol{\theta}^k - \bar{\boldsymbol{\theta}}^k]^2, \ \{\lambda^k, \bar{\boldsymbol{\theta}}^k\} = \begin{cases} \{1, 0\} & \forall \theta = \alpha \\ \left\{\frac{1}{\sqrt{.5}}, 3\right\} & \theta = \beta \quad k = \tilde{g} \\ \left\{\frac{1}{\sqrt{.1}}, 3\right\} & \theta = \beta \quad k = r^* \\ \{1, 1\} & \text{otherwise}. \end{cases} \tag{37}$$

Equation (37) centers the parameters α and β around zero and one, implying that we start from the assumption that the information provided by the observed series is unbiased. The only exception is the coefficient $\beta^{\tilde{g}}$, which we center around 3 to acknowledge that the observed value of industrial production might be more volatile than our latent series.

3.2.1 Term structure model

We estimate the parameters of Equation (20) using a quasi-maximum likelihood procedure. Specifically, we fix the parameters $\boldsymbol{\mu}^{\mathbb{P}}$ and $\Phi^{\mathbb{P}}$ of Equation (17) and we estimate the parameters $\boldsymbol{\mu}^{\chi}$, Φ^{χ} and Σ^{p} by fitting the term structure of the ILS rates. As in Joslin, Singleton, and Zhu (2011), the starting point for the estimation of Σ^{p} are the state-space estimates. We maximize the likelihood function by combining simulated annealing and a simplex procedure.

4 Empirical Results

The following two subsections focus on decomposing ILS rates in expected and IRP components. We present the results for a model with three lags. Appendix A.3 reports the parameter estimates, plots of the stochastic trends, and the fit of the measurement equations. The t-statistics and confidence intervals are based on 500 bootstrapped samples obtained as in Stoffer and Wall (1991). We focus our analysis on the 2-year (short-term) and 10-year (long-term) contracts, which are maturities also analyzed by studies employing ILS data (Haubrich, Pennacchi, and Ritchken 2012; Berardi and Plazzi 2019) and TIPS data (Breach, D'Amico, and Orphanides 2020).

The third subsection describes the variance decomposition of the ILS, EC, and IRP according to the different groups of shocks using the full sample. In the fourth subsection, we focus on the main drivers during four episodes of interest.

4.1 Expected Component

Figure 2a (b) reports the evolution of the 2-year (10-year) ILS rates and its EC and ILS decompositions for the US (sub panel A) and the EA (sub panel B). Table A6 reports the

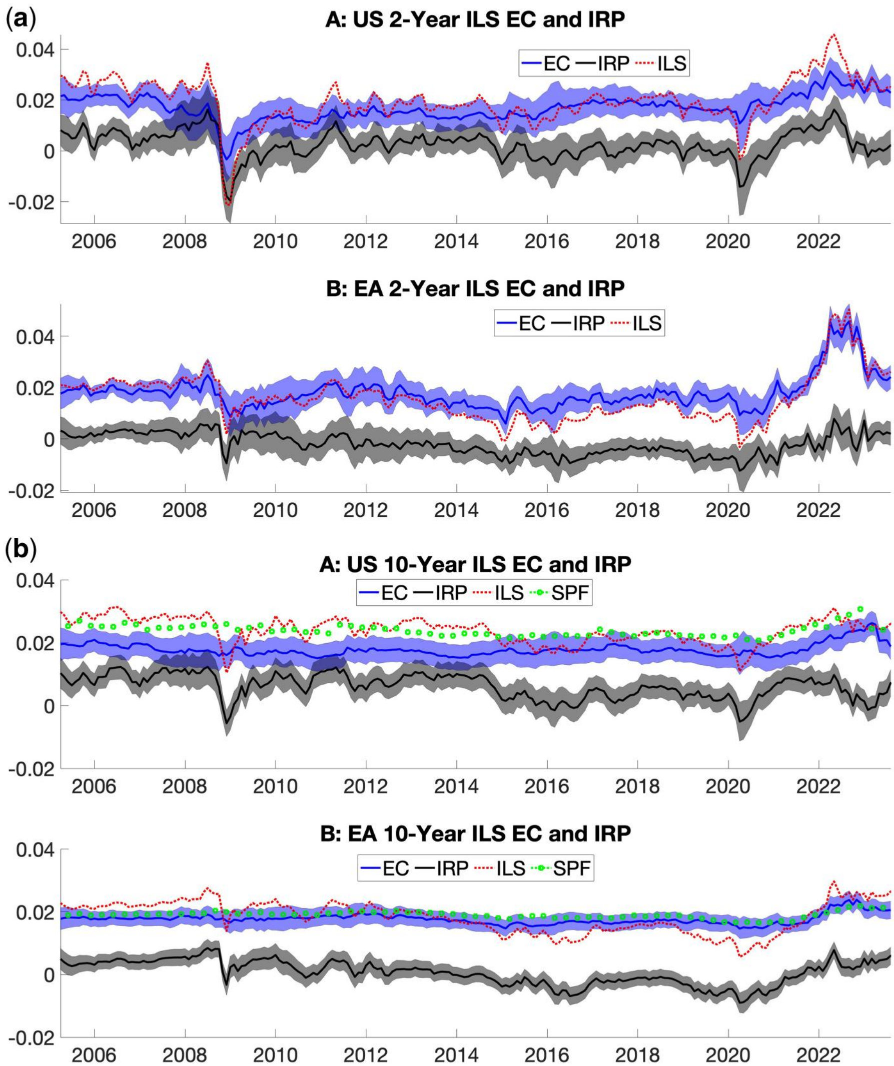

Figure 2. Two-year and 10-year rates, expected and inflation risk premia components.

Notes: (a) For the 2-year ILS rates, the subpanel (A) and (B) of this figure report the US and EA ILS rates, and their decompositions in EC (blue) and IRP (grey). (b) This panel reproduces similar plots to (a) but for a 10-year ILS rate. Shaded areas indicate the 95% confidence intervals. Green circles represent the 5-year (10-year) inflation forecasts from the EA (US) SPF.

summary statistics of the EC and IRP components for the full sample and the pre-January 2022 sample.

As in Berardi and Plazzi (2019), we find that US short-term expectations—proxied by the EC of the ILS rates—are relatively volatile, fluctuating between −34 and 313 bps. Furthermore, on average, they are comparable to other EC measures extracted from ILS contracts. For example, over the period 2009–2015, our average two-year expected inflation is comparable to the estimates of Fleckenstein, Longstaff, and Lustig (2017) (only 10 bps smaller), even though their measure is roughly twice as volatile as ours. The most volatile periods relate to the GFC, the COVID-19 period, and the Ukraine war, reflecting the market's fear of the possibility of a prolonged recession (GFC and early stage of COVID-19) and the fear of high inflation brought about by supply chain bottlenecks, geopolitical tensions, and large fiscal stimulus packages (last part of COVID-19 crisis and Ukraine war). The EC of the short-term ILS rates for the EA are considerably more volatile than their US counterparts, with a 15 bps higher standard deviation. The 2-year EA EC fluctuates between 60 and 459 bps, and a significant portion of this variation can be attributed to the recent Ukraine invasion, which has caused the standard deviation of the short-term EC in the EA to nearly double, rising from 37 to 65 bps.

Turning to longer-term EC, two observations stand out. First, they are substantially less volatile than their short-term counterparts. For example, in the EA (US), the standard deviation of the 10-year EC is roughly one-quarter (one-third) of the 2-year analog. This finding is also documented in the literature. For example, Fleckenstein, Longstaff, and Lustig (2017) find that in the US, the standard deviation of the 10-year EC of ILS is a fraction (one-third) of the 2-year counterpart while Berardi and Plazzi (2019) document that long-term expectations are quite stable and fluctuate within the range 115–215 bps, which is slightly lower than our 150–262 bps estimates (which include the recent period of high inflation). Second, despite a more volatile EC in the short end, the long-term EA EC is less volatile than its US counterpart. Potentially, the latter observation might reflect that, in contrast to the dual mandate of the FED, the ECB's mandate focuses primarily on inflation stabilization. However, the EA ILS rates hover below the EC between 2013 and 2021 due to persistently negative risk premia. Therefore, a careful reading of inflation swaps is needed to draw firm conclusions regarding the anchoring of inflation expectations.

Figure 2b also shows the SPF long-term survey observations used in the estimation (green dots). The EA SPF expectations fall within the confidence bands of the EC estimates. However, the 10-year EC is persistently below the survey observations for the US. This feature is not unique to our model, as the 10-year EC estimates of Haubrich, Pennacchi, and Ritchken (2012) for the US are similar to our 10-year EC series and thus also persistently below the survey data.[13]

4.2 Inflation Risk Premium

The short-term US IRP fluctuates between −197 and 163 bps, with the most abrupt changes happening around Lehman's default when it dropped by roughly 320 bps over four months. It matches Fleckenstein, Longstaff, and Lustig (2017) averages—and references therein—but is three times more volatile. It was positive before the GFC, a period characterized by relatively stable inflation and growth perspectives; see Figure 3. The deterioration in (expected) economic conditions, associated with the low inflation outlook, contributed to the flip in the sign of the IRP at the end of 2008. Similarly to Haubrich, Pennacchi, and Ritchken (2012), the risk premia fluctuated in negative territory until the end of 2010. However, in the latter part of the period, negative premia might have been brought about by the post-crisis recovery of the US economy and the return of inflation to its pre-crisis levels. In the succeeding period, the US inflation risk premia turned negative

[13] Their series is available at https://www.clevelandfed.org/indicators-and-data/inflation-expectations.

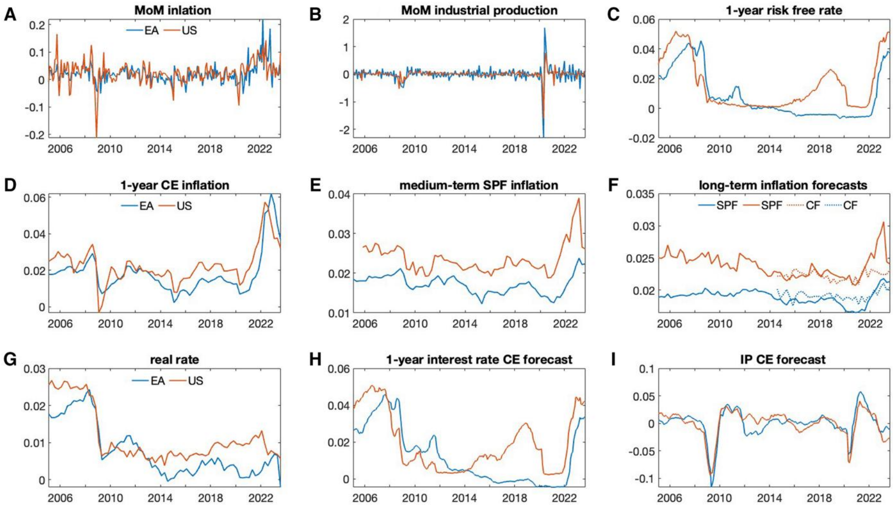

Figure 3. Observable variables.

Notes: The top panels report the observed series of month-on-month inflation, industrial production growth ((A) and (B), respectively), and the one-year risk-free rate (C). The middle panels depict the inflation surveys. Panel (D) plots the monthly Consensus Economics (CE) one-year-ahead inflation forecast, panel E reports the two (five) year inflation Survey of Professional Forecast (SPF) for the EA (US), and panel (F) depicts (i) the five (10) year inflation forecast from the SPF for the EA (US), and (ii) the long-term (above five years) CE inflation forecasts. Finally, the bottom panels show the quarterly Holston et al. (2017) natural rate of interest (G), and the monthly CE one-year risk-free rate (H) and industrial production (I) forecast. In each panel, blue lines refer to EA data, while red lines are for US data.

around the 2010s oil price glut, which was combined with the mini-recession of 2015, and from mid-2019 until the end of 2020, a period characterized first by less-than-expected economic growth, and then by the COVID-19 crisis. Our results corroborate the findings of Fleckenstein, Longstaff, and Lustig (2017), Haubrich, Pennacchi, and Ritchken (2012), and Camba-Mendez and Werner (2017), who suggest that (i) around the GFC and the oil price glut, the probability of deflation might be as high as 30% for a 2-year horizon (contract), that (ii) the two-year IRP was substantially negative during the GFC, and that (iii) short-term US IRP can be negative for a prolonged period.

The short-term EA IRP varies between −124 and 78 bps and is, on average, negative. It was positive prior to the GFC but turned negative after its inception. In contrast to the US case, EA risk premia did not fall as much and quickly reverted to positive values. Berardi and Plazzi (2019) show that volatility and IRP are negatively related in periods of crisis or market-wide distress. Hence, the quick flip in the sign of EA risk premia is potentially due to the region's lower volatility of inflation (and its projections). From the beginning of 2013, risk premia started to decline and turned negative for a prolonged period until the 2022 Russian invasion of Ukraine. This period coincides with the so-called EA lowflation, during which a benign economic recovery coincided with inflation remaining persistently weak and below the ECB's then-prevailing inflation aim of below, but close to, 2% in the medium term (Stevens and Wauters 2021). The decline in risk premia intensified around the oil price glut and, after a partial recovery in 2018, reached their lowest levels when the COVID-19 crisis erupted.

In both areas, the 10-year maturity IRP is, on average, positive and more volatile than the EC. This result is standard for the US and is found elsewhere in the literature; see, for example, Fleckenstein, Longstaff, and Lustig (2017) and Haubrich, Pennacchi, and Ritchken (2012) for studies using alternative approaches with a comparable dataset.

Our IRP is 30% more volatile than the estimates of Fleckenstein, Longstaff, and Lustig (2017), and its mean lies between their 22 bps estimates and the 67 bps and 115 bps averages of Chernov and Mueller (2012) and Campbell and Viceira (2001), respectively. The longer-term IRP is negative only in three episodes—the GFC, the oil glut, and the COVID-19 crisis (in the EA, the IRP was also negative before that period)—and for only a short period. For the EA, over the period January 2012–July 2018, Böninghausen, Kidd, and de Vincent-Humphreys (2018) find a similar pattern for the market-based long-term IRP, attributing most of the fall in (5-year) ILS rates (in 5 years) over the period from 2014 to 2016 to deflation risk. The higher volatility implies that most of the variation in longer-term ILS rates, see Table 1, comes from IRP's movements.[14] From the COVID-19 crisis onward, we observe an increase in the long-term EA IRP. This finding relates to Mouabbi, Renne, and Tschopp (2023), who document a rising contribution of adverse supply shocks to inflation and a significant decrease in the correlation between inflation and economic activity. Applying the terminology of Cieslak and Pflueger (2023), this could be considered as the beginning of a period of "bad inflation," wherein supply-driven inflation dynamics result in positive long-term (bond) risk premiums.

4.3 Deterministic Components

Figure A3 reports the deterministic components for the US and EA 2-year and 10-year ILS rates and their decomposition in EC and IRP parts, respectively, based on our model specification (top panels) and a "latent" model with only the three principal components as risk factors (bottom panels). Giannone, Lenza, and Primiceri (2019) point out that it is important to discipline the long-run behavior of vector autoregression models. Otherwise, the deterministic components might, implausibly, capture a large share of the variation in the observed time series. Our model can produce deterministic components of the ILS rates that are not influenced by its low-frequency movements. This contrasts with the results obtained from a three-factor latent model, whereby for the EA, the deterministic components of the ILS rates display oscillating behavior, especially for the 2-year ILS rates, to explain a share of the low-frequency variability observed in the data. The problem is pronounced for models estimated before the Ukraine crisis (dotted lines).

4.4 Variance Decomposition

We report the variance decomposition of the 2-year and 10-year US (EA) ILS rates, its expected and IRP components, in panels A–C and D–F of Table 2, respectively. We aggregate the shocks in four groups: (i) Σ_{PCs}, which refers to the *principal components* of the yield curve and is the sum of level, slope, and curvature innovations; (ii) Σ_{Gl}, namely the sum of oil and *global* real activity innovations; (iii) Σ_{Loc} representing the sum of the *local* macro variable innovations (month-on-month inflation, month-on-month industrial production growth, and the one-year risk-free rate); and (iv) the sum of *long-run* innovations (inflation and real rate trends), labeled Σ_{LR}.

The stochastic variation in ILS rates in both regions is influenced by short-term and long-term economic innovations, with varying contributions depending on the contract's maturity and the frequency of the movements being analyzed. We first focus on the short-term ILS contracts. Our model reveals that transitory economic developments drive high-frequency movements (12-month horizon) in the 2-year ILS rates. However, the extent to which local and international factors contribute to these movements differs between regions. In the US, international innovations account for most stochastic variation (35% in total, compared to 27% for local economic factors). In contrast, in the EA, local economic innovations contribute to approximately half of the total movements.

[14] Decomposing the ILS volatility into the EC, IRP, and correlation component highlight that for the EA (US) 72% (90%) of 10-year ILS volatility is due to the IRP component.

Table 2. Error variance decomposition

		US											
		A: ILS level				B: Expected component				C: Inflation risk premia			
Maturity	Horizon	Σ_{PCs}	Σ_{Gl}	Σ_{Loc}	Σ_{LR}	Σ_{PCs}	Σ_{Gl}	Σ_{Loc}	Σ_{LR}	Σ_{PCs}	Σ_{Gl}	Σ_{Loc}	Σ_{LR}
2-year	12M	0.32	0.35	0.27	0.06	0.20	0.10	0.38	0.32	0.20	0.64	0.14	0.03
2-year	120M	0.13	0.17	0.13	0.57	0.04	0.09	0.08	0.79	0.18	0.65	0.13	0.03
10-year	12M	0.29	0.31	0.13	0.27	0.02	0.04	0.05	0.90	0.28	0.51	0.06	0.15
10-year	120M	0.05	0.06	0.03	0.86	0.00	0.01	0.01	0.98	0.21	0.58	0.06	0.14
		EA											
		D: ILS level				E: Expected component				F: Inflation risk premia			
Maturity	Horizon	Σ_{PCs}	Σ_{Gl}	Σ_{Loc}	Σ_{LR}	Σ_{PCs}	Σ_{Gl}	Σ_{Loc}	Σ_{LR}	Σ_{PCs}	Σ_{Gl}	Σ_{Loc}	Σ_{LR}
2-year	12M	0.26	0.23	0.50	0.01	0.12	0.22	0.61	0.05	0.49	0.16	0.30	0.04
2-year	120M	0.23	0.23	0.43	0.10	0.09	0.17	0.46	0.28	0.46	0.22	0.25	0.07
10-year	12M	0.49	0.18	0.32	0.01	0.07	0.11	0.33	0.49	0.60	0.15	0.21	0.05
10-year	120M	0.38	0.17	0.22	0.23	0.02	0.02	0.07	0.90	0.56	0.19	0.18	0.07

Notes: For a 2-year and 10-year US (EA) ILS contract panels A–C (D–F) report the error variance decomposition of the ILS rate level, the EC, and inflation risk premia, respectively. Innovations are grouped as: (i) Σ_{PCs}, which is the sum of level, slope, and curvature innovations; (ii) Σ_{Gl}, namely the sum of oil and global real activity innovations; (iii) Σ_{Loc} representing the sum of the local macro variable innovations (month-on-month inflation, month-on-month industrial production growth, and one-year risk-free rate); and (iv) the sum of long-term innovations (long-run inflation and real rate trends), labeled Σ_{LR}. The error variance decompositions refer to horizons of 12 months (first row of each maturity) and 120 months (second row of each maturity).

When analyzing low-frequency movements of ILS rates (10-year horizon), a more fragmented picture emerges across the two regions. In the US, short-term rates are less influenced by transitory economic innovations, which explain only about one-third of the total variation, while long-term innovations play a crucial role, accounting for half of the total ILS low-frequency variation. Conversely, in the EA, transitory economic innovations account for 66%, while long-term factors contribute only 10% of the total variation.

A closer examination of the decomposition of ILS rates reveals three key patterns. First, in the US, international factors primarily impact ILS rates through the IRP channel, while in the EA, local factors are the most important economic drivers of both the IRP and the EC. In the US, two-thirds of IRP variation can be attributed to international innovations, while for the EA, local factors account for around one-quarter (half) of the IRP (EC) fluctuations.

Second, long-term economic trends significantly impact the low-frequency movement of the EC, particularly in the US, where stochastic trend innovations explain three-quarters of the total variation, compared to one-quarter in the EA. Finally, local economic innovations are mainly associated with variation in the EC of short-term ILS rates, especially in the EA, where they account for at least 60% of total variation.

The variance decomposition of the 10-year maturity indicates that, for both areas, long-run innovations play a larger role in explaining the movements of longer-term ILS rates compared to short-term contracts. This result is particularly pronounced for low-frequency movements, whereby innovations in stochastic trends account for more than 80 and 20% of the variation in US and EA rates, respectively. The EC is the channel mostly linked to long-term innovations, with more than 98 and 90% of the US and EA variation attributed to long-run innovations.

Finally, we find that spanned factor shocks are a key component of the IRP variations, accounting for roughly one-fifth (half) of the US (EA) high- and low-frequency IRP movements.

Next, we quantify how innovations in short and long-term variables influence ILS rates throughout recent episodes of interest.

4.5 Historical Decomposition of Four Episodes

In this section, we decompose the 2-year and 10-year US and EA ILS rates and their expected and IRP components, respectively, around four historical episodes: (i) the GFC, which is characterized by an initial drop in ILS rates during the second half of 2008—brought about by lower inflation expectations linked to the global downturn caused by the crisis—and then by a rebound in the first half of the subsequent year; (ii) the oil price glut period, roughly spanning from mid-2014 to the end of 2015, and that is distinguished by downward pressure on inflation expectations; (iii) the COVID-19 era, ranging from early 2020 until the end of 2021. The period is marked by a sharp decline in ILS rates followed by an increasing trend that continued until the end of the period; and (iv) the Russian invasion in Ukraine, spanning throughout 2022, whereby ILS rates initially surged to historically high values, see Figure 1.

The second column of Panel A1 and A2 of Table 3 reports the changes in the 2-year ILS during the periods indicated in the first column. These date ranges mark the most pronounced movements in ILS rates during these episodes. The subsequent column decomposes the ILS changes in (i) the sum of PC innovations (column 3), (ii) the contribution of international innovations (columns 4–5), (iii) the part related to local economic innovations (columns 6–8), and (iv) the long-run innovations (columns 9–10). Panels B and C are structured similarly but refer to the EC and IRP parts of the ILS changes, respectively. Table 4 repeats the analysis for the 10-year maturity.

4.5.1 Great financial crisis

Focusing on the 2-year contract during the GFC, Table 3 indicates that, in both regions, innovations to international factors, and in particular oil prices, played a key role in the stochastic movements of the ILS rates and its EC and IRP components. From June 2008 to December 2008, the US ILS rates experienced a drop of 563 bps, which is split between a 36% decrease in the EC (−204 bps) and a 64% decline in the IRP (−359 bps). The decrease in oil prices accounted for about 30% of the ILS rate drop, that is, −177 bps. The other major macroeconomic factors pushing down those rates were the drop in inflation that is not accounted for by other factors (−124 bps), global (commodity) demand (−65 bps), and industrial production (−60 bps). The decomposition of ILS rates reveals that, while international innovations work mostly via the IRP channel—76% of their total contribution relates to this channel—local macroeconomic factors evenly affected both the EC and the IRP. In the following months, from January 2009 to December 2009, ILS rates bounced back by 316 bps, and the relative contribution of individual innovations nearly mirrored the previous period but with a switch in sign. Oil price increases were the leading drivers of ILS and IRP surges (by 134 and 79 bps, respectively), followed by other transitory economic factors, particularly global demand and inflation. Over the same period, the 2-year EA ILS rates exhibit half of the variation compared to their US counterparts, with most of this volatility attributable to IRP movements. As in the US, oil price innovations were (one of) the dominant factors, accounting for 32% of the initial ILS drop and 46% of their subsequent increase. In contrast to the US case, our model indicates the importance of interest rates in explaining ILS movements in the EA. Interest rates were brought to very low levels at the end of the period and accounted for 38% of the drop in ILS rates. Monetary policy

Table 3. Detailed historical decomposition: 2-year ILS

US										EA									
Panel A1: Level										**Panel A2: Level**									
GFC	y_2	*PCs*	g^w	p^o	π	g	i	π^*	r^*	GFC	y_2	*PCs*	g^w	p^o	π	g	i	π^*	r^*
2008.06 2008.12	−563	−143	−65	−177	−124	−60	−13	6	14	2008.06 2008.11	−284	−60	−50	−91	2	19	−108	2	2
2008.11 2009.06	316	66	40	134	57	−3	6	−1	17	2008.11 2009.06	149	100	21	68	−50	11	−13	4	9
Oil Glut	y_2	*PCs*	g^w	p^o	π	g	i	π^*	r^*	Oil Glut	y_2	*PCs*	g^w	p^o	π	g	i	π^*	r^*
2014.06 2014.12	−148	−19	2	−125	0	−4	2	5	−9	2014.06 2015.01	−92	−16	−6	−91	30	−7	4	−3	−2
2014.12 2015.04	105	24	−5	93	9	−1	−8	1	−8	2015.01 2015.04	113	26	−2	80	25	−10	−2	−1	−1
2015.04 2015.09	−92	−39	4	−36	−1	−11	−3	−3	−4	2015.04 2015.08	−74	−25	8	−47	−10	−1	4	−2	−1
COVID-19	y_2	*PCs*	g^w	p^o	π	g	i	π^*	r^*	COVID-19	y_2	*PCs*	g^w	p^o	π	g	i	π^*	r^*
2019.12 2020.03	−208	3	−18	−184	10	0	−21	−2	4	2019.12 2020.03	−133	−28	−11	−83	−11	−7	8	−2	1
2020.03 2020.08	218	−10	20	183	4	13	−3	−1	12	2020.03 2020.08	104	47	23	66	−7	−9	−13	−1	0
2020.10 2021.05	168	71	19	70	25	−13	−1	−3	0	2020.10 2021.03	136	−2	9	67	58	1	5	−2	−1
2021.08 2021.11	67	70	−10	−30	40	−10	1	6	−1	2021.04 2021.12	145	56	3	−17	105	3	−3	−1	0
Ukraine war	y_2	*PCs*	g^w	p^o	π	g	i	π^*	r^*	Ukraine war	y_2	*PCs*	g^w	p^o	π	g	i	π^*	r^*
2021.12 2022.04	118	60	0	42	−33	10	34	12	−8	2021.12 2022.04	219	81	14	40	39	8	35	2	0
2022.04 2022.09	−224	−128	−30	−71	−20	−25	47	32	−27	2022.08 2023.01	−276	−191	−4	−42	−51	−5	19	6	−7
Panel B1: EC										**Panel B2: EC**									
GFC	y_2^{EC}	*PCs*	g^w	p^o	π	g	i	π^*	r^*	GFC	y_2^{EC}	*PCs*	g^w	p^o	π	g	i	π^*	r^*
2008.06 2008.12	−204	−67	−24	−35	−58	−34	−10	13	12	2008.06 2008.11	−135	−26	−1	−74	−10	4	−41	13	−1
2008.11 2009.06	137	28	21	55	32	4	8	−17	5	2008.11 2009.06	23	25	−4	46	−47	−3	−10	5	11
Oil Glut	y_2^{EC}	*PCs*	g^w	p^o	π	g	i	π^*	r^*	Oil Glut	y_2^{EC}	*PCs*	g^w	p^o	π	g	i	π^*	r^*
2014.06 2014.12	−26	−9	2	−25	1	3	2	9	−11	2014.06 2015.01	−62	−1	−2	−79	29	−2	2	−7	−1
2014.12 2015.04	20	14	−1	41	2	−14	−8	−6	−8	2015.01 2015.04	103	11	1	57	38	−1	−1	−2	−1
2015.04 2015.09	−20	−19	6	−4	3	0	−2	−3	−1	2015.04 2015.08	−64	−14	−1	−39	−10	0	2	0	−1

(continued)

Table 3. (continued)

Panel B1: EC

COVID-19	y_2^{EC}	PCs	g^w	p^o	π	g	i	π^*	r^*
2019.12 2020.03	−57	1	−6	−44	3	4	−19	1	3
2020.03 2020.08	94	−6	8	66	2	18	−1	−5	11
2020.10 2021.05	49	33	2	5	13	−8	0	7	−3
2021.08 2021.11	50	33	−8	−12	21	7	1	10	−2
Ukraine war	y_2^{EC}	PCs	g^w	p^o	π	g	i	π^*	r^*
2021.12 2022.04	35	32	−6	1	−17	0	32	11	−18
2022.04 2022.09	−39	−57	−9	−19	−4	−14	37	42	−15

Panel B2: EC

COVID-19	y_2^{EC}	PCs	g^w	p^o	π	g	i	π^*	r^*
2019.12 2020.03	−86	−4	3	−81	−1	−3	3	−4	0
2020.03 2020.08	13	7	−1	50	−36	−2	−5	2	−2
2020.10 2021.03	56	−17	2	54	20	1	2	−6	0
2021.04 2021.12	129	18	1	−6	115	−1	−1	2	0
Ukraine war	y_2^{EC}	PCs	g^w	p^o	π	g	i	π^*	r^*
2021.12 2022.04	122	39	5	37	11	5	13	14	−1
2022.08 2023.01	−223	−142	0	−28	−74	0	12	16	−6

Panel C1: IRP

GFC	y_2^{IRP}	PCs	g^w	p^o	π	g	i	π^*	r^*
2008.06 2008.12	−359	−77	−41	−142	−66	−26	−3	−7	2
2008.11 2009.06	179	37	19	79	25	−7	−1	15	12
Oil Glut	y_2^{IRP}	PCs	g^w	p^o	π	g	i	π^*	r^*
2014.06 2014.12	−121	−10	−1	−101	−1	−7	0	−4	2
2014.12 2015.04	85	10	−4	52	7	13	0	7	0
2015.04 2015.09	−72	−20	−2	−32	−3	−11	−1	0	−3
COVID-19	y_2^{IRP}	PCs	g^w	p^o	π	g	i	π^*	r^*
2019.12 2020.03	−151	2	−12	−140	6	−4	−2	−3	1
2020.03 2020.08	124	−4	12	117	2	−5	−2	4	1
2020.10 2021.05	120	38	17	65	12	−5	0	−10	3
2021.08 2021.11	16	37	−1	−18	19	−17	0	−4	0
Ukraine war	y_2^{IRP}	PCs	g^w	p^o	π	g	i	π^*	r^*
2021.12 2022.04	83	28	6	41	−16	10	3	1	10
2022.04 2023.01	−185	−71	−21	−52	−17	−11	10	−11	−12

Panel C2: IRP

GFC	y_2^{IRP}	PCs	g^w	p^o	π	g	i	π^*	r^*
2008.06 2008.11	−148	−34	−48	−17	12	15	−67	−12	3
2008.11 2009.06	125	75	25	22	−3	13	−3	−1	−2
Oil Glut	y_2^{IRP}	PCs	g^w	p^o	π	g	i	π^*	r^*
2014.06 2015.01	−30	−15	−4	−12	1	−5	2	4	0
2015.01 2015.04	10	15	−3	23	−14	−9	−1	0	0
2015.04 2015.08	−10	−11	9	−8	0	−1	2	−2	0
COVID-19	y_2^{IRP}	PCs	g^w	p^o	π	g	i	π^*	r^*
2019.12 2020.03	−47	−24	−14	−3	−10	−5	5	2	1
2020.03 2020.08	92	40	24	16	29	−8	−8	−3	1
2020.10 2021.03	80	15	7	12	38	0	3	4	−1
2021.04 2021.12	17	38	3	−11	−10	4	−2	−3	0
Ukraine war	y_2^{IRP}	PCs	g^w	p^o	π	g	i	π^*	r^*
2021.12 2022.04	97	42	9	4	28	3	22	−12	1
2022.08 2023.01	−54	−49	−4	−14	24	−5	7	−10	−2

Notes: For a 2-year US (EA) ILS contract, the second column of panels A1–C1 (A2–C2) report the changes in the stochastic part of the ILS rate level (y_2), the EC (y_2^{EC}), and inflation risk premia (y_2^{IRP}), respectively. The first columns report the periods over which the changes are computed, while the third columns reports the total changes in the PCs. The remaining columns list the changes in the other variables, labeled following the notation of Section 2.2. We drop the time subscript to ease the notation.

Table 4. Detailed historical decomposition: 10-year ILS

US										EA									
Panel A1: Level										**Panel A2: Level**									
GFC	y_{10}	*PCs*	g^w	p^o	π	g	i	π^*	r^*	GFC	y_{10}	*PCs*	g^w	p^o	π	g	i	π^*	r^*
2008.06 2008.12	−167	−16	−36	−69	−34	−22	−9	8	11	2008.06 2008.11	−136	−22	−28	−36	−2	17	−68	1	1
2008.11 2009.06	142	51	19	41	17	4	2	−4	13	2008.11 2009.06	90	64	1	30	−16	16	−15	4	6
Oil Glut	y_{10}	*PCs*	g^w	p^o	π	g	i	π^*	r^*	Oil Glut	y_{10}	*PCs*	g^w	p^o	π	g	i	π^*	r^*
2014.06 2014.12	−66	−15	−1	−47	0	−3	1	7	−8	2014.06 2015.01	−53	−17	−6	−34	10	−4	3	−3	−1
2014.12 2015.04	15	−4	−5	29	3	2	−4	1	−7	2015.01 2015.04	33	12	−5	31	9	−10	−1	−1	−1
2015.04 2015.09	−44	−12	0	−18	−2	−5	−2	−4	−2	2015.04 2015.08	−24	−5	2	−18	−2	−1	3	−2	−1
COVID-19	y_{10}	*PCs*	g^w	p^o	π	g	i	π^*	r^*	COVID-19	y_{10}	*PCs*	g^w	p^o	π	g	i	π^*	r^*
2019.12 2020.03	−86	3	−11	−70	2	0	−12	−2	3	2019.12 2020.03	−64	−22	−10	−33	−2	−1	5	−1	0
2020.03 2020.08	92	9	9	64	1	3	−4	−1	10	2020.03 2020.08	47	37	6	29	−4	−10	−8	−2	−1
2020.10 2021.05	59	26	12	26	6	−6	−1	−2	−2	2020.10 2021.03	65	20	3	25	17	0	3	−2	0
2021.08 2021.11	18	19	−2	−11	9	−5	1	8	−1	2021.04 2021.12	66	32	10	−8	30	6	−2	−2	0
Ukraine war	y_{10}	*PCs*	g^w	p^o	π	g	i	π^*	r^*	Ukraine war	y_{10}	*PCs*	g^w	p^o	π	g	i	π^*	r^*
2021.12 2022.04	37	−4	3	18	−10	5	19	13	−8	2021.12 2022.04	88	30	3	15	14	4	22	1	0
2022.04 2023.01	−69	−61	−15	−23	−5	−6	31	35	−23	2022.08 2023.01	−26	−11	−3	−14	−16	0	19	5	−5
Panel B1: EC										**Panel B2: EC**									
GFC	y_{10}^{EC}	*PCs*	g^w	p^o	π	g	i	π^*	r^*	GFC	y_{10}^{EC}	*PCs*	g^w	p^o	π	g	i	π^*	r^*
2008.06 2008.12	−8	−13	3	3	−13	−7	−3	19	3	2008.06 2008.11	−20	−5	−1	−15	−2	0	−10	13	−1
2008.11 2009.06	−2	6	0	9	7	1	2	−28	1	2008.11 2009.06	9	5	0	9	−9	−1	−3	5	4
Oil Glut	y_{10}^{EC}	*PCs*	g^w	p^o	π	g	i	π^*	r^*	Oil Glut	y_{10}^{EC}	*PCs*	g^w	p^o	π	g	i	π^*	r^*
2014.06 2014.12	12	−2	1	2	0	1	1	13	−3	2014.06 2015.01	−17	0	0	−16	6	0	0	−7	0
2014.12 2015.04	−7	3	2	7	0	−3	−2	−10	−2	2015.01 2015.04	19	2	0	11	8	0	0	−2	0
2015.04 2015.09	−5	−4	0	2	1	0	−1	−4	0	2015.04 2015.08	−13	−3	0	−8	−2	0	0	0	0

(continued)

Table 4. (continued)

Panel B1: EC

COVID-19	y_{10}^{EC}	PCs	g^w	p^o	π	g	i	π^*	r^*
2019.12 2020.03	3	0	2	1	1	1	−5	4	1
2020.03 2020.08	5	−1	−1	7	0	4	0	−7	3
2020.10 2021.05	15	7	−2	−5	3	−2	0	14	−1
2021.08 2021.11	24	7	−1	−1	5	2	0	14	−1
Ukraine war	y_{10}^{EC}	PCs	g^w	p^o	π	g	i	π^*	r^*
2021.12 2022.04	14	7	0	−3	−4	0	9	11	−6
2022.04 2023.01	43	−12	1	0	−1	−3	10	51	−4

Panel B2: EC

COVID-19	y_{10}^{EC}	PCs	g^w	p^o	π	g	i	π^*	r^*
2019.12 2020.03	−20	−1	0	−16	0	0	1	−4	0
2020.03 2020.08	3	1	0	10	−7	0	−1	2	−1
2020.1 2021.03	6	−4	0	11	4	0	0	−6	0
2021.04 2021.12	27	3	0	−1	23	0	0	2	0
Ukraine war	y_{10}^{EC}	PCs	g^w	p^o	π	g	i	π^*	r^*
2021.12 2022.04	35	8	1	7	2	1	3	14	0
2022.08 2023.01	−33	−30	0	−6	−15	0	3	16	−2

Panel C1: IRP

GFC	y_{10}^{IRP}	PCs	g^w	p^o	π	g	i	π^*	r^*
2008.06 2008.12	−159	−3	−39	−72	−21	−15	−6	−11	7
2008.11 2009.06	145	45	19	32	10	3	0	24	12
Oil Glut	y_{10}^{IRP}	PCs	g^w	p^o	π	g	i	π^*	r^*
2014.06 2014.12	−78	−13	−1	−49	0	−3	1	−6	−5
2014.12 2015.04	21	−6	−7	23	3	4	−2	11	−5
2015.04 2015.09	−39	−8	−1	−20	−3	−5	−2	0	−2
COVID-19	y_{10}^{IRP}	PCs	g^w	p^o	π	g	i	π^*	r^*
2019.12 2020.03	−89	3	−13	−71	2	−1	−6	−5	2
2020.03 2020.08	87	10	10	57	1	0	−4	6	7
2020.10 2021.05	44	19	14	31	3	−4	−1	−16	−1
2021.08 2021.11	−6	13	−1	−10	4	−7	0	−6	−1
Ukraine war	y_{10}^{IRP}	PCs	g^w	p^o	π	g	i	π^*	r^*
2021.12 2022.04	22	−10	3	21	−6	5	10	2	−2
2022.04 2023.01	−111	−50	−16	−23	−4	−4	21	−16	−19

Panel C2: IRP

GFC	y_{10}^{IRP}	PCs	g^w	p^o	π	g	i	π^*	r^*
2008.06 2008.11	−116	−17	−27	−21	0	17	−57	−12	1
2008.11 2009.06	81	59	2	21	−7	17	−12	−1	3
Oil Glut	y_{10}^{IRP}	PCs	g^w	p^o	π	g	i	π^*	r^*
2014.06 2015.01	−36	−18	−6	−19	4	−4	3	4	−1
2015.01 2015.04	14	9	−5	20	1	−10	−1	0	0
2015.04 2015.08	−11	−2	2	−10	0	−1	3	−2	0
COVID-19	y_{10}^{IRP}	PCs	g^w	p^o	π	g	i	π^*	r^*
2019.12 2020.03	−44	−21	−10	−18	−2	−1	5	2	0
2020.03 2020.08	44	36	6	19	3	−10	−7	−3	0
2020.1 2021.03	59	24	2	14	13	0	2	4	−1
2021.04 2021.12	39	28	10	−7	7	6	−1	−4	0
Ukraine war	y_{10}^{IRP}	PCs	g^w	p^o	π	g	i	π^*	r^*
2021.12 2022.04	53	22	2	8	12	3	18	−13	0
2022.08 2023.01	7	18	−3	−9	−1	0	16	−11	−3

Notes: For a 10-year US (EA) ILS contract, the second column of panels A1 to C1 (A2–C2) report the changes in the stochastic part of the ILS rate level (y_{10}), the EC (y_{10}^{EC}), and inflation risk premia (y_{10}^{IRP}), respectively. The first columns report the periods when the changes are computed, while the third columns report the total changes in the PCs. The remaining columns list the changes in the other variables, labeled following the notation of Section 2.2. We drop the time subscript to ease the notation.

weighed on inflation compensation by accounting for more than 45% of the decline of IRP in the aftermath of Lehman's collapse.

4.5.2 Oil price glut

During the oil price glut episodes of the mid-2010s, in both areas, the 2-year ILS rates oscillated over three consecutive periods, each time by about 100 basis points in absolute value. During the first phase of the oil glut, from June 2014 to December 2014, oil price innovations drove these fluctuations, accounting—mainly via the IRP channel—for more than 80 and 95% of the decrease in US and EA ILS rates. In the second phase of the glut, the increase in oil prices contributed to the surge of ILS rates, mainly due to an increase in both the EC and IRP. Finally, during the last decline in oil prices, the drop in ILS was also linked to PC innovations, that is, to factors not captured by our model.

Our results concur with findings obtained using alternative models and data. For example, using US household survey data, Kilian and Zhou (2022) show that oil (gasoline) price drops were the key driver of short-term (one-year) expectations in these two phases. This result largely aligns with our EC historical decomposition for the US. In addition, Elliott et al. (2015) find a statistically significant effect of oil prices on both US and EA area short-term (3-year) ILS over the period January 2009 to July 2015 while Conflitti and Cristadoro (2018) document for the EA an increase in the correlation between oil price changes and 1-year ILS rates for the period July 2014–January 2016. Finally, using zero coupon and year-on-year inflation options, Galati et al. (2018) highlight that over January 2013–December 2015, 2-year and 10-year (see infra) deflation risk became more sensitive to oil price changes concerning previous periods.

4.5.3 COVID-19 pandemic

During the first phase of COVID-19, US (EA) ILS rates initially dropped by 208 bps (133 bps) following the collapse of the global economy induced by the widespread restrictive sanitary measures. Subsequently, they rose by 218 bps (104 bps). The separation of these variations in EC and IRP components highlights that US ILS variations are chiefly linked to the latter channel, with the cumulative effects of global demand and, to a larger extent, oil price innovation accounting for virtually all the variation of the ILS rates (and its decomposition). In the EA, the initial drop of ILS rates was linked to both EC (65% of total variation) and IRP fluctuations, while their subsequent increase was related to the IRP channel. Innovations related to oil price changes were key drivers of EC and IRP changes, followed by global demand, inflation, and PCs innovations.

In both areas, the post-COVID-19 period, which ranges from late 2020 to late 2021, is characterized by two surges in ILS rates, reflecting the concerns linked to rising global demand coupled with supply chain disruptions and the rebound of energy prices. The first increase started in October 2020, when US and EA rates climbed by 168 bps and 136 bps, mainly due to an increase in the IRP, which represented roughly three-quarters (three-fifths) of the total variation. In the US, international innovations were still the primary source of variation in ILS rates, even if (i) their relative importance dropped to roughly 50% of the total variation and (ii) they spawned variation only in the IRP component. Kilian and Zhou (2022) show that, over the same period, the recovery of gasoline prices accounted for an increase in one-year inflation expectations of around 70 bps and were unimportant for the five-year horizon. Our results partly confirm this finding as for the EC of the 1-year contract, the impact of oil innovations is about 35 bps but quickly dies out for longer maturities, that is, 2 and 10 years; see also Table 4. In the EA, the climb in ILS rates is mainly linked to local inflation dynamics and oil innovations. Interestingly, our model reveals that part of the inflation-induced increase was linked to a rise in the EC, potentially indicating that market participants believed in a short-term, temporary climb in inflation, chiefly linked to oil and inflation innovations.

The second increase in the US and EA began in August and April 2021, respectively, and was mainly associated with changes in the EC of ILS rates, especially in the EA. In this region, the surge in the ILS rate was associated with the expectation of higher inflation rates in the following years due to concerns related to the near-term outlook of inflation dynamics—the cumulated contribution of inflation innovations accounted for virtually all the variation in the EC.

4.5.4 Ukraine invasion

The last episode we inspect covers the recent Russian invasion of Ukraine. Contrary to the previous episodes, we document that (i) EA rates show greater variability than the US peers and (ii) EA ILS changes are driven by fluctuations in the EC, while the reverse is true for the US, which fluctuated, mainly following variations in IRP. For the US, the detailed decomposition highlights that most fluctuations in ILS rates (and its EC and IRP components) relate to PCs and oil innovations. The EA decomposition reveals that the initial surge in observed rates is linked—at various degrees—to the cumulative contribution of PCs, inflation, oil, and interest rates innovation. In contrast, the subsequent decrease in ILS rates is mainly related to PC innovations. Domestic and oil factors play a more important role, consistent with the EA being more exposed to this conflict.

4.5.5 10-year ILS decomposition

The historical decomposition of the 10-year contract (Table 4) highlights the following points. First, in line with the analysis of Section 4.1 in both areas, the ILS rates are less volatile than their shorter-maturity counterparts. The larger swings in ILS rates occur in both regions during the GFC. The decline and the subsequent recovery of ILS rates were largely associated with transitory innovations, that is, international—oil in particular—local inflation and demand (with the addition of interest rates for the EA). These changes were largely connected to risk repricing, signaling that the market did foresee only a marginal impact of those innovations on long-term expected inflation—in line with the relatively stable pattern, especially for the EA, of the stochastic trend inflation over the period, see Figure A1. The relatively stable pattern of the long-term EC partly contrasts the findings of Galati, Poelhekke, and Zhou (2011), who measure (daily) long-run inflation expectations as the one-year forward rate in nine years, cleaned for liquidity premium, and find that during the GFC, this measure of inflation expectations became more sensitive to news about inflation and other domestic macroeconomic variables.

Second, before mid-2021, the changes in ILS rates were mostly due to variations in the IRP channel, with the EC varying for a maximum of 20 bps. After this date, corresponding to the last three periods of Table 4, our model signals more pronounced movements in long-term ILS-based inflation expectations, which substantially contributed to ILS rate fluctuations. Using average estimates of two dynamic term structure models and EA data, Neri et al. (2022) show that, after mid-2021, there has been an increase in both the IRP and genuine inflation expectations of the 5-year ILS rates in 5 years, corroborating our results. For the US, our model attributes these developments chiefly to innovations to the stochastic trend for inflation, which increased in the last part of the sample, see Figure A1. Conversely, for the EA, the higher variation of the EC component of the 10-year rates is mainly linked to transitory inflation innovations and, for the inflation spike, also to PC innovations and variations in the stochastic trend of inflation. Such concerns about deanchoring of inflation expectations explain the forceful reaction of central banks to the inflation spike during 2022.

In summary, the historical decomposition reveals the importance of international factors in shaping the dynamics of these rates during specific periods. While similarities exist between the US and the EA, there are also key differences, such as the importance of local innovations in shaping the EA market. The analysis also highlights that for short-term maturities (2 years), economic innovations work through the EC and the IRP channels, pointing out that they are incorporated in marked-based inflation expectation measures, even when refined from risk compensation. For longer maturities, the variations in ILS rates are mostly related to the IRP channel, even if we observe that during the recent COVID-19 crisis, the EC channel is becoming more volatile, partly due to the contribution of economic innovations.

5 Conclusions

Measuring inflation expectations and their anchoring to the inflation target is important for monetary policy at central banks. One source of information on expectations is ILS rates, as quoted on financial markets. These series are available at a high frequency and reflect market activity. However, they are not a pure measure of inflation expectations because they contain a risk premium component.

Our article proposes a new ATSM to decompose ILS rates into genuine inflation expectations and risk premia. By incorporating both transitory macroeconomic factors and long-term economic drivers, such as trend inflation and equilibrium real interest rates, our model offers a more comprehensive understanding of the dynamics influencing ILS rates in the US and EA.

Our main findings are the following. First, the IRP is a significant driver of ILS rates, especially at longer maturities. The risk premium was positive in the US and EA before the GFC but then switched signs and became negative for a prolonged period in the EA.

Second, the relative importance of short versus long-term (macroeconomic) shocks depends on the ILS contract's maturity, the region, and whether one considers high or low-frequency movements. Short-term macroeconomic factors are important in both regions, but the composition differs. While international factors like oil prices are key in the US, local factors (like inflation and interest rates) are relatively more important in the EA. Innovations to the stochastic trends play a larger role in explaining longer-term ILS rates, especially their ECs. Inflation risk premia are importantly driven by spanned factor shocks that only affect the principal components of the ILS term structure.

Third, focusing on recent key global events like the COVID-19 pandemic and the Russian invasion of Ukraine, we find that international factors, particularly oil prices, were crucial drivers of ILS rates by impacting both the inflation EC and risk premia.

Overall, our results recommend a careful reading of ILS rates for policy decisions: shifts in the inflation risk premia can give a misleading impression of the degree of inflation expectations anchoring, and the dynamics of ILS rates are determined by different factors depending on the contract's maturity, the region, and whether high or low-frequency movements are of interest.

Our article identifies short-term macroeconomic shocks by making assumptions on their transmission speed to different macroeconomic and financial variables (Cholesky identification). One avenue for future work is to apply a different shock identification scheme within our framework.

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Appendix A

A.1 Data Codes

Table A1. Data

Variable	Source (and *codes*)
Price inflation	ECB (*ICP.M.U2.Y.X02200.3.INX*) FRED (*CPIAUCSL*)
Industrial Production Index	ECB (*STS.M.I8.Y.PROD.NS0020.4.000*) FRED (*INDPRO*)
Risk-free rates	FRED (*DTB1YR*)[1]
Real oil price	ECB (*FM.M.GB.EUR.4F.CY.EUCRBRDT.HSTA*, *ICP.M.U2.Y.000000.3.INX*) FRED (*OILPRICE*, *CPIAUCSL*)
Index of global real economic activity	Dallas Fed (*igrea*)

Note: ECB data for the euro area are retrieved from the ECB data portal at https://data.ecb.europa.eu/. Data for the US are taken from the FRED database at https://fred.stlouisfed.org/. The global measure of real economic activity is available at: https://www.dallasfed.org/research/igrea.

[1] Since the series is discontinued until June 2008, prior to that date, we use the one-year rate obtained from Liu and Wu (2021).

A.2 Regression Evidence of Unspanned Macro Variation and Unspanned Macro Risk

In our modeling setting, we impose that three ILS principal components entirely explain the cross-section of ILS rates. Also, the global, country-specific, and long-term trends contain information about the future evolution of ILS rates above the information spanned by the principal components. This modeling choice overwhelmingly simplifies the estimation of the model, as the estimation of the risk factor dynamics, see Equation (17), can be (partially) separated from the yield curve fitting, see Equation (24). In this section, we provide regression evidence supporting our modeling choice. We follow Sections 4.2 and 5.1 of Bauer and Rudebusch (2017) and report model-free regression evidence of unspanned macro variation and unspanned macro risk.

Investigating for unspanned macro variation is equivalent to asking: Do ILS principal components embed all the information in the global, country-specific, and long-term trends? In other words, can we write each of those variables as a perfect linear combination of the ILS principal components? Table A2 depicts the adjusted R^2 of regressing each macroeconomic variable on (i) all the first three principal components of the ILS rates (first row) and (ii) on each principal component separately (remaining rows). The stochastic trends are proxied by the (i) one-year ahead CE inflation expectation and (ii) the difference between the one-year risk-free rate and the one-year ahead CE inflation expectation. Focusing on the first row, for the US, except for the log of oil prices and inflation expectations, all the adjusted R^2 are below 50%. For the EA, ILS factors capture substantial variation of the risk-free rate and inflation expectations. For the other variables, the explanatory power is roughly equal to or smaller than 50%. The information contained in the level of the ILS rates links mostly to the one-year risk-free rate, inflation expectation, and inflation rate. This aligns with intuition, as all these variables are directly or indirectly linked to inflation.

We now turn to the model-free analysis related to unspanned macro risk. We analyze the ex-post excess returns entering at time t in a n-period ILS contract as the fixed leg counterpart and then closing this position after twelve months (at $t+12$) by taking the opposite position on an *n-12* contract. We label this quantity $\mathcal{R}^n_{t,t+12}$ and we write it, in annual terms, as:

$$\mathcal{R}^n_{t,t+12} = Y_{n,t} - \frac{n-12}{n} Y_{n-12,t+12} - \frac{12}{n}\pi_{\tau,\tau+12}, \tag{A.1}$$

where $Y_{n,t}$ and $Y_{n-12,t+12}$ are ILS rates, and $\pi_{\tau,\tau+1}$ is the floating payment related to the strategy, i.e., the one-year inflation between the τ and $\tau+1$, and $\tau<t$ accounts for the lagged indexation. The first column of Table A3 reports the adjusted R^2 of regressing

Table A2. Unspanned macro variation

	US: adjusted R^2								EA: adjusted R^2						
	p^o_t	g^w_t	π_t	g_t	i_t	π^*_t	r^*_t		p^o_t	g^w_t	π_t	g_t	i_t	π^*_t	r^*_t
Joint	0.59	0.36	0.47	0.08	0.24	0.60	0.04	Joint	0.51	0.51	0.41	−0.01	0.64	0.76	0.60
PC1	0.18	0.23	0.39	0.06	0.21	0.49	0.00	PC1	0.33	0.28	0.30	0.00	0.40	0.67	0.00
PC2	0.36	0.13	0.04	0.00	0.00	0.05	0.05	PC2	0.01	0.05	0.11	0.00	0.15	0.08	0.45
PC3	0.04	0.00	0.04	0.03	0.03	0.05	0.00	PC3	0.17	0.17	0.00	0.00	0.08	0.00	0.15

Notes: The first row reports the adjusted R^2 of regressing each macroeconomic variable on the first three principal components of the ILS rates. The last three rows report the adjusted R^2 of performing the regressions only with the first (second row), second (third row), or third (fourth row) principal component.

Table A3. Unspanned macro risk

	US realized returns										
	$\mathbf{p}_t$	Global variables			Local variables				Stochastic trends		
ILS	R^2_{adj}	p^o_t	g^w_t	ΔR^2_{adj}	π_t	g_t	i_t	ΔR^2_{adj}	π^*_t	r^*_t	ΔR^2_{adj}
1-year	0.04	1.24	−0.45	0.21	−1.81	−0.10	0.07	0.07	0.37	0.14	0.01
pval		0.00	0.10		0.00	0.48	0.67		0.22	0.43	
2-year	0.10	1.98	−0.27	0.21	−1.66	−0.20	0.13	0.03	0.47	0.20	0.01
pval		0.00	0.47		0.03	0.36	0.57		0.28	0.40	
10-year	0.20	3.33	−0.70	0.16	−2.17	−0.13	−0.03	0.00	−0.09	0.07	−0.01
pval		0.00	0.27		0.10	0.72	0.94		0.90	0.88	
	EA realized returns										
	$\mathbf{p}_t$	Global variables			Local variables				Stochastic trends		
ILS	R^2_{adj}	p^o_t	g^w_t	ΔR^2_{adj}	π_t	g_t	i_t	ΔR^2_{adj}	π^*_t	r^*_t	ΔR^2_{adj}
1-year	0.15	0.37	−0.56	0.08	−1.68	−0.02	0.39	0.11	1.23	0.40	0.11
pval		0.09	0.04		0.01	0.43	0.06		0.01	0.07	
2-year	0.19	0.85	−0.94	0.09	−1.78	−0.05	0.62	0.07	2.09	0.58	0.09
pval		0.02	0.06		0.04	0.33	0.06		0.02	0.07	
10-year	0.29	2.28	−1.75	0.14	−1.79	−0.04	−0.20	0.00	1.13	−0.22	0.00
pval		0.00	0.02		0.24	0.74	0.71		0.50	0.67	

Notes: The first column reports the adjusted R^2 of regressing one-year realized excess returns on the first three principal components of the ILS rates. The remaining columns report the coefficients and the increase in adjusted R^2 by adding to $\mathbf{p}_t$ either (i) the level of the global variables, columns two to four, (iii) the local variables, columns five to eight, and (iv) two proxies for the stochastic trends, last three columns. In all cases, the global variables have been standardized to have zero mean and standard deviation of 1%. P-values are based on Newey–West standard errors with automatic lags selection.

$\mathcal{R}^n_{t,t+12}$ on the first three principal ILS components, $\mathbf{p}_t$. The remaining columns report the coefficients and the increase in adjusted R^2 by adding to $\mathbf{p}_t$ either (i) the level of the global variables, columns two to four, (iii) the local variables, columns five to eight, and (iv) two proxies for the stochastic trends, last three columns (in all cases the global variables have been standardized to have zero mean and standard deviation of 1%. P-values are based on Newey–West standard errors with automatic lags selection).

Although PCs tend to better explain realized returns of contracts with longer maturities for both regions, the explanatory power of the current shape of the ILS term structure is stronger for the EA than for the US. In the US, global variables help to explain ILS returns, especially for shorter-term contracts. Oil prices are at the origin of these results, with the oil coefficient significant at any confidence level. The explanatory power of global factors is less pronounced in the EA but tends to increase with maturity. Contrary to the US, both factors are equally responsible for the increase in adjusted R-squared. In both areas, the local variables exhibit explanatory power associated with short-term contracts. However, the origins of this explanatory power differ between the US and the EA. In the US, the explanatory power is connected to inflation, whereas in the EA, it is mostly tied to short-term interest rates. Finally, turning to stochastic trends, we report that they are mostly linked to returns on EA short-term contracts (via the inflation trend), while there is little explanatory power in the US market (we also perform the predictive regressions by using the two-year moving average of our proxy for stochastic trends as explanatory variables. The explanatory power of stochastic trends, in particular the one linked to inflation, increases both for the EA and the US.).

Overall, the predictive power for ILS returns tends to improve when international, local, or stochastic trend variables are added, yet the effects depend on the region and contract maturity.

A.3 Estimation Results

We obtained the error bands and t-statistics via bootstrapped standard errors. We use 500 draws, obtained via the bootstrap procedure for state space models of Stoffer and Wall (1991), adapted to account for missing observations.

A.3.1 Parameter Estimates

Table A4. US parameter estimates

parameter	times	mode	tStat	parameter	times	mode	tStat	parameter	times	mode	tStat	parameter	times	mode	tStat	parameter	times	mode	tStat
σ^{g^w,g^w}	1	0.23	17.25	$\phi_1^{g^w,g^w}$	1	1.09	34.18	$\phi_2^{g^w,i}$	10^{-2}	−0.10	−1.43	$\phi_3^{p^o,g^w}$	1	−0.08	−3.95	μ^l	10^{-2}	−0.48	−4.33
σ^{g^w,p^o}	1	0.01	1.32	$\phi_1^{g^w,p^o}$	1	0.05	2.54	$\phi_2^{g^w,l}$	10^{-2}	−0.45	−1.77	$\phi_3^{p^o,p^o}$	1	0.17	6.81	μ^s	10^{-2}	0.66	6.46
$\sigma^{g^w,\pi}$	10^{-2}	0.57	1.59	$\phi_1^{g^w,\pi}$	10^{-2}	0.48	0.67	$\phi_2^{g^w,s}$	10^{-2}	−0.29	−2.42	$\phi_3^{p^o,\pi}$	1	0.02	1.97	μ^c	10^{-2}	0.13	2.75
$\sigma^{g^w,g}$	10^{-2}	0.04	0.07	$\phi_1^{g^w,g}$	1	0.02	0.90	$\phi_2^{g^w,c}$	10^{-2}	0.13	2.30	$\phi_3^{p^o,g}$	1	−0.03	−1.52	π_0^*	1	0.03	9.41
$\sigma^{g^w,i}$	10^{-4}	0.92	0.58	$\phi_1^{g^w,i}$	10^{-2}	0.08	1.51	$\phi_2^{p^o,g^w}$	1	−0.07	−3.08	$\phi_3^{p^o,i}$	10^{-2}	0.22	1.96	r_0^*	1	0.03	2.61
$\sigma^{g^w,l}$	10^{-2}	0.16	1.94	$\phi_1^{g^w,l}$	10^{-2}	0.21	1.09	$\phi_2^{p^o,p^o}$	1	−0.19	−8.58	$\phi_3^{p^o,l}$	10^{-2}	0.44	1.20	σ^{ip}	10^{-2}	0.18	2.50
$\sigma^{g^w,s}$	10^{-2}	0.02	0.68	$\phi_1^{g^w,s}$	10^{-2}	0.20	2.08	$\phi_2^{p^o,\pi}$	1	−0.09	−4.21	$\phi_3^{p^o,s}$	10^{-2}	−0.19	−0.93	$\sigma^{ce,ip}$	1	0.03	36.39
$\sigma^{g^w,c}$	10^{-4}	0.17	0.17	$\phi_1^{g^w,c}$	10^{-2}	−0.08	−1.73	$\phi_2^{p^o,g}$	1	−0.06	−3.54	$\phi_3^{p^o,c}$	10^{-2}	−0.06	−0.58	$\sigma_{lr}^{ce,\pi}$	10^{-2}	0.50	4.70
σ^{p^o,p^o}	1	0.11	7.80	$\phi_1^{p^o,g^w}$	1	0.12	4.60	$\phi_2^{p^o,i}$	10^{-2}	0.10	0.60	$\phi_3^{\pi,\pi}$	1	0.21	8.35	$\sigma_{lr}^{ce,i}$	10^{-2}	0.85	16.26
$\sigma^{p^o,\pi}$	1	0.01	3.46	$\phi_1^{p^o,p^o}$	1	0.99	37.78	$\phi_2^{p^o,l}$	1	−0.01	−2.14	$\phi_3^{\pi,g}$	1	0.03	1.49	$\sigma_{sr}^{ce,\pi}$	10^{-2}	0.67	9.63
$\sigma^{p^o,g}$	10^{-2}	0.43	0.64	$\phi_1^{p^o,\pi}$	1	0.07	4.20	$\phi_2^{p^o,s}$	10^{-2}	0.31	1.18	$\phi_3^{\pi,i}$	10^{-2}	0.86	2.52	$\sigma_{sr}^{spf,\pi}$	10^{-2}	0.16	5.35
$\sigma^{p^o,i}$	10^{-2}	0.05	2.08	$\phi_1^{p^o,g}$	1	0.12	5.07	$\phi_2^{p^o,c}$	10^{-2}	−0.19	−1.64	$\phi_3^{\pi,l}$	10^{-2}	−0.54	−0.52	$\sigma_{lr}^{spf,\pi}$	10^{-2}	0.08	4.64
$\sigma^{p^o,l}$	10^{-2}	0.48	5.11	$\phi_1^{p^o,i}$	10^{-2}	−0.31	−2.66	$\phi_2^{\pi,\pi}$	1	−0.16	−3.86	$\phi_3^{\pi,s}$	10^{-2}	−0.33	−0.58	$\sigma_{sr}^{ce,i}$	10^{-2}	0.08	6.05
$\sigma^{p^o,s}$	10^{-2}	−0.01	−0.57	$\phi_1^{p^o,l}$	10^{-2}	0.27	0.70	$\phi_2^{\pi,g}$	1	0.02	0.91	$\phi_3^{\pi,c}$	10^{-2}	0.20	0.71	$\sigma^{r,hlw}$	10^{-2}	0.43	16.25
$\sigma^{p^o,c}$	10^{-2}	0.02	1.47	$\phi_1^{p^o,s}$	10^{-4}	−0.98	−0.05	$\phi_2^{\pi,i}$	1	−0.02	−3.17	$\phi_3^{g,\pi}$	10^{-2}	0.76	0.61	$\alpha^{r,hlw}$	1	0.01	10.02
$\sigma^{\pi,\pi}$	1	0.03	11.83	$\phi_1^{p^o,c}$	10^{-2}	0.30	2.63	$\phi_2^{\pi,l}$	1	−0.04	−3.45	$\phi_3^{g,g}$	1	0.17	4.90	β^{ip}	1	4.85	12.20
$\sigma^{\pi,g}$	10^{-4}	−0.89	−1.08	$\phi_1^{\pi,\pi}$	1	0.42	8.77	$\phi_2^{\pi,s}$	10^{-2}	0.93	1.39	$\phi_3^{g,i}$	10^{-2}	0.20	1.42	$\beta^{r,hlw}$	1	0.37	7.80
$\sigma^{\pi,i}$	10^{-2}	0.04	1.98	$\phi_1^{\pi,g}$	1	0.21	4.48	$\phi_2^{\pi,c}$	10^{-2}	−0.53	−1.52	$\phi_3^{g,l}$	10^{-2}	0.83	1.74	$\alpha_{sr}^{ce,\pi}$	10^{-2}	0.06	0.99
$\sigma^{\pi,l}$	10^{-2}	0.22	3.69	$\phi_1^{\pi,i}$	1	0.01	2.97	$\phi_2^{g,\pi}$	1	0.01	0.80	$\phi_3^{g,s}$	10^{-2}	0.04	0.16	$\alpha_{lr}^{ce,\pi}$	1	0.02	99.43
$\sigma^{\pi,s}$	10^{-2}	−0.06	−1.90	$\phi_1^{\pi,l}$	1	0.06	4.63	$\phi_2^{g,g}$	1	0.04	0.91	$\phi_3^{g,c}$	10^{-2}	0.01	0.11	$\alpha_{lr}^{ce,i}$	1	0.02	26.13
$\sigma^{\pi,c}$	10^{-2}	0.03	2.24	$\phi_1^{\pi,s}$	10^{-2}	−0.64	−1.03	$\phi_2^{g,i}$	10^{-2}	−0.31	−1.52	$\phi_3^{i,\pi}$	1	−0.07	−2.70	$\beta_{sr}^{ce,i}$	1	1.01	245.97
$\sigma^{g,g}$	10^{-2}	0.08	8.17	$\phi_1^{\pi,c}$	10^{-2}	−0.02	−0.06	$\phi_2^{g,l}$	1	0.01	1.73	$\phi_3^{i,g}$	1	−0.03	−1.56	λ_1	1	1.00	17122.57

(continued)

Table A4. (continued)

parameter	times	mode	tStat	parameter	times	mode	tStat	parameter	times	mode	tStat	parameter	times	mode	tStat	parameter	times	mode	tStat
$\sigma^{g,i}$	10^{-4}	0.44	0.31	$\phi_1^{g,\pi}$	1	−0.02	−0.78	$\phi_2^{g,s}$	10^{-2}	0.15	0.52	$\phi_3^{i,i}$	1	−0.15	−5.70	λ_2	1	0.96	108.16
$\sigma^{g,l}$	10^{-2}	−0.06	−1.03	$\phi_1^{g,g}$	1	0.11	1.31	$\phi_2^{g,c}$	10^{-2}	0.36	2.51	$\phi_3^{i,l}$	1	0.02	1.33	λ_3	1	0.93	29.21
$\sigma^{g,s}$	10^{-2}	−0.01	−0.48	$\phi_1^{g,i}$	10^{-2}	0.18	1.03	$\phi_2^{i,\pi}$	1	0.02	1.13	$\phi_3^{i,s}$	10^{-2}	0.39	0.29	ρ_0	1	0.12	5.64
$\sigma^{g,c}$	10^{-2}	0.01	0.71	$\phi_1^{g,l}$	1	0.01	2.47	$\phi_2^{i,g}$	1	−0.04	−2.06	$\phi_3^{i,c}$	1	0.02	1.88				
$\sigma^{i,i}$	10^{-2}	0.18	11.93	$\phi_1^{g,s}$	10^{-2}	−0.40	−−1.52	$\phi_2^{i,i}$	1	0.16	11.07	$\phi_1^{l,l}$	1	0.81	27.13				
$\sigma^{i,l}$	10^{-2}	0.11	2.32	$\phi_1^{g,c}$	10^{-2}	−0.10	−0.81	$\phi_2^{i,l}$	1	−0.03	−1.60	$\phi_1^{l,s}$	1	−0.02	−1.15				
$\sigma^{i,s}$	10^{-2}	0.02	0.75	$\phi_1^{i,\pi}$	1	0.31	7.55	$\phi_2^{i,s}$	10^{-2}	−0.11	−0.08	$\phi_1^{l,c}$	10^{-2}	−0.92	−0.87				
$\sigma^{i,c}$	10^{-2}	−0.01	−0.91	$\phi_1^{i,g}$	1	0.03	1.41	$\phi_2^{i,c}$	1	−0.01	−1.25	$\phi_1^{s,l}$	1	0.06	1.84				
$\sigma^{l,l}$	10^{-2}	0.48	16.86	$\phi_1^{i,i}$	1	0.94	37.87	$\phi_3^{g^w,g^w}$	1	0.06	2.63	$\phi_1^{s,s}$	1	0.87	25.74				
$\sigma^{l,s}$	10^{-2}	0.02	0.69	$\phi_1^{i,l}$	1	0.10	4.39	$\phi_3^{g^w,p^o}$	1	−0.01	−0.76	$\phi_1^{s,c}$	1	0.01	0.72				
$\sigma^{l,c}$	10^{-4}	−1.00	−0.87	$\phi_1^{i,s}$	1	0.06	2.65	$\phi_3^{g^w,\pi}$	1	0.03	4.64	$\phi_1^{c,l}$	1	−0.09	−2.84				
$\sigma^{s,s}$	10^{-2}	0.25	14.66	$\phi_1^{i,c}$	1	−0.02	−0.85	$\phi_3^{g^w,g}$	1	−0.02	−1.68	$\phi_1^{c,s}$	1	0.38	5.58				
$\sigma^{s,c}$	10^{-2}	−0.06	−4.15	$\phi_2^{g^w,g^w}$	1	−0.22	−8.66	$\phi_3^{g^w,i}$	10^{-4}	0.98	0.19	$\phi_1^{c,c}$	1	0.58	13.71				
$\sigma^{c,c}$	10^{-2}	0.12	12.25	$\phi_2^{g^w,p^o}$	1	−0.03	−1.47	$\phi_3^{g^w,l}$	10^{-2}	0.34	1.79								
σ^{π^*,π^*}	10^{-2}	0.07	4.28	$\phi_2^{g^w,\pi}$	1	−0.03	−3.72	$\phi_3^{g^w,s}$	10^{-2}	0.14	1.38								
σ^{r^*,r^*}	10^{-2}	0.25	3.00	$\phi_2^{g^w,g}$	10^{-4}	−0.24	0.00	$\phi_3^{g^w,c}$	10^{-2}	−0.06	−1.21								

Notes: The parameter names (columns 1, 4, 8, 12, and 16, respectively) closely follow the notation of Sections 2 and 3. Each parameter estimate (mode) is multiplied by the factor listed under the accompanying "times" column.

Table A5. EA parameter estimates

parameter	times	mode	tStat	parameter	times	mode	tStat	parameter	times	mode	tStat	parameter	times	mode	tStat	parameter	times	mode	tStat
σ^{g^w,g^w}	1	0.23	16.57	$\phi_1^{g^w,g^w}$	1	1.11	47.63	$\phi_2^{g^w,i}$	10^{-2}	−0.08	−1.34	$\phi_3^{p^o,g^w}$	1	−0.08	−4.80	μ^l	10^{-2}	−0.25	−6.39
σ^{g^w,p^o}	10^{-2}	0.92	1.77	$\phi_1^{g^w,p^o}$	1	0.04	3.06	$\phi_2^{g^w,l}$	10^{-2}	−0.32	−1.42	$\phi_3^{p^o,p^o}$	1	0.09	4.87	μ^s	10^{-2}	0.41	8.13
$\sigma^{g^w,\pi}$	10^{-2}	0.67	3.02	$\phi_1^{g^w,\pi}$	10^{-2}	−0.13	−0.27	$\phi_2^{g^w,s}$	10^{-2}	−0.02	−0.24	$\phi_3^{p^o,\pi}$	1	0.03	4.23	μ^c	10^{-2}	0.43	10.51
$\sigma^{g^w,g}$	10^{-2}	−0.11	−0.18	$\phi_1^{g^w,g}$	1	0.03	1.34	$\phi_2^{g^w,c}$	10^{-2}	0.06	1.43	$\phi_3^{p^o,g}$	10^{-2}	−0.63	−0.39	π_0^*	1	0.02	18.83
$\sigma^{g^w,i}$	10^{-4}	0.03	1.14	$\phi_1^{g^w,i}$	10^{-2}	0.06	1.39	$\phi_2^{p^o,g^w}$	1	−0.14	−7.29	$\phi_3^{p^o,i}$	10^{-2}	0.13	1.26	r_0^*	1	0.02	5.21
$\sigma^{g^w,l}$	10^{-2}	0.11	1.68	$\phi_1^{g^w,l}$	10^{-2}	0.14	0.81	$\phi_2^{p^o,p^o}$	1	−0.29	−14.50	$\phi_3^{p^o,l}$	10^{-2}	0.27	0.70	σ^{ip}	10^{-2}	0.25	11.02
$\sigma^{g^w,s}$	10^{-2}	0.01	0.79	$\phi_1^{g^w,s}$	10^{-2}	0.07	1.04	$\phi_2^{p^o,\pi}$	1	−0.05	−5.96	$\phi_3^{p^o,s}$	10^{-2}	0.05	0.30	$\sigma^{ce,ip}$	1	0.03	73.03
$\sigma^{g^w,c}$	10^{-2}	−0.02	−1.49	$\phi_1^{g^w,c}$	10^{-2}	−0.02	−0.64	$\phi_2^{p^o,g}$	1	−0.06	−3.68	$\phi_3^{p^o,c}$	10^{-2}	0.04	0.47	$\sigma_{lr}^{ce,\pi}$	10^{-2}	0.08	76.39
σ^{p^o,p^o}	1	0.08	11.27	$\phi_1^{p^o,g^w}$	1	0.13	6.68	$\phi_2^{p^o,i}$	10^{-4}	0.03	0.00	$\phi_3^{\pi,\pi}$	1	0.25	12.63	$\sigma_{lr}^{ce,i}$	10^{-2}	0.01	18.45
$\sigma^{p^o,\pi}$	1	0.01	5.91	$\phi_1^{p^o,p^o}$	1	1.15	42.36	$\phi_2^{p^o,l}$	10^{-2}	−0.50	−0.91	$\phi_3^{\pi,g}$	1	0.05	3.05	$\sigma_{sr}^{ce,\pi}$	10^{-2}	0.72	30.75
$\sigma^{p^o,g}$	1	0.03	5.15	$\phi_1^{p^o,\pi}$	1	0.02	1.34	$\phi_2^{p^o,s}$	10^{-2}	−0.37	−1.56	$\phi_3^{\pi,i}$	10^{-2}	−0.10	−0.37	$\sigma_{sr}^{spf,\pi}$	10^{-2}	0.17	12.20
$\sigma^{p^o,i}$	10^{-2}	0.02	1.61	$\phi_1^{p^o,g}$	1	0.08	4.12	$\phi_2^{p^o,c}$	10^{-2}	−0.06	−0.54	$\phi_3^{\pi,l}$	1	−0.02	−2.01	$\sigma_{lr}^{spf,\pi}$	10^{-2}	0.08	259.81
$\sigma^{p^o,l}$	10^{-2}	0.30	4.89	$\phi_1^{p^o,i}$	10^{-2}	−0.11	−1.09	$\phi_2^{\pi,\pi}$	1	0.19	8.63	$\phi_3^{\pi,s}$	1	0.02	5.09	$\sigma_{sr}^{ce,i}$	10^{-2}	0.08	13.71
$\sigma^{p^o,s}$	10^{-2}	−0.05	−3.60	$\phi_1^{p^o,l}$	10^{-2}	0.03	0.07	$\phi_2^{\pi,g}$	10^{-2}	−0.39	−0.24	$\phi_3^{\pi,c}$	10^{-2}	−0.28	−1.17	$\sigma^{r,blw}$	10^{-2}	0.31	13.72
$\sigma^{p^o,c}$	10^{-4}	−0.49	−0.61	$\phi_1^{p^o,s}$	10^{-2}	0.28	1.52	$\phi_2^{\pi,i}$	10^{-2}	−0.14	−0.33	$\phi_3^{g,\pi}$	1	−0.02	−2.06	$\alpha^{r,blw}$	10^{-2}	0.86	10.86
$\sigma^{\pi,\pi}$	1	0.02	8.82	$\phi_1^{p^o,c}$	10^{-2}	0.13	1.36	$\phi_2^{\pi,l}$	1	0.05	4.49	$\phi_3^{g,g}$	1	0.20	7.13	β^{ip}	1	4.91	35.35
$\sigma^{\pi,g}$	1	−0.02	−2.59	$\phi_1^{\pi,\pi}$	1	0.17	5.66	$\phi_2^{\pi,s}$	1	−0.02	−3.02	$\phi_3^{g,i}$	10^{-2}	−0.01	−0.28	$\beta^{r,blw}$	1	0.41	13.49
$\sigma^{\pi,i}$	10^{-4}	0.54	0.35	$\phi_1^{\pi,g}$	1	0.21	9.50	$\phi_2^{\pi,c}$	10^{-2}	−0.07	−0.27	$\phi_3^{g,l}$	10^{-4}	−0.30	−0.01	$\alpha_{sr}^{ce,\pi}$	10^{-2}	0.07	1.52
$\sigma^{\pi,l}$	10^{-2}	0.14	2.39	$\phi_1^{\pi,i}$	10^{-2}	0.21	0.54	$\phi_2^{g,\pi}$	10^{-2}	−0.39	−0.40	$\phi_3^{g,s}$	10^{-2}	−0.20	−2.52	$\alpha_{lr}^{ce,\pi}$	10^{-2}	0.04	3.15
$\sigma^{\pi,s}$	10^{-2}	−0.02	−1.37	$\phi_1^{\pi,l}$	1	0.08	6.03	$\phi_2^{g,g}$	1	0.07	2.71	$\phi_3^{g,c}$	10^{-2}	0.06	1.48	$\alpha_{lr}^{ce,i}$	1	0.01	17.85
$\sigma^{\pi,c}$	10^{-4}	0.79	0.86	$\phi_1^{\pi,s}$	1	−0.03	−5.27	$\phi_2^{g,i}$	10^{-2}	−0.03	−0.40	$\phi_3^{i,\pi}$	1	−0.23	−11.00	$\beta_{sr}^{ce,i}$	1	1.01	372.71
$\sigma^{g,g}$	1	0.14	9.87	$\phi_1^{\pi,c}$	10^{-2}	0.75	2.71	$\phi_2^{g,l}$	10^{-2}	−0.35	−1.47	$\phi_3^{i,g}$	1	−0.03	−2.05	λ_1	1	1.00	1751.91
$\sigma^{g,i}$	10^{-2}	0.03	2.14	$\phi_1^{g,\pi}$	10^{-2}	0.58	0.54	$\phi_2^{g,s}$	10^{-2}	−0.23	−2.31	$\phi_3^{i,i}$	10^{-2}	−0.05	−0.03	λ_2	1	0.98	112.05

(continued)

Table A5. (continued)

parameter	times	mode	tStat	parameter	times	mode	tStat	parameter	times	mode	tStat	parameter	times	mode	tStat	parameter	times	mode	tStat
$\sigma^{g,l}$	10^{-2}	−0.05	−0.82	$\phi_1^{g,g}$	1	−0.11	−4.33	$\phi_2^{g,c}$	10^{-2}	0.08	1.62	$\phi_3^{i,l}$	1	0.03	2.04	λ_3	1	0.87	44.74
$\sigma^{g,s}$	10^{-4}	−0.15	−0.08	$\phi_1^{g,i}$	10^{-4}	0.97	0.14	$\phi_2^{i,\pi}$	1	−0.09	−4.37	$\phi_3^{i,s}$	10^{-2}	−0.63	−0.57	ρ_0	1	0.03	3.24
$\sigma^{g,c}$	10^{-4}	−0.07	−0.07	$\phi_1^{g,l}$	10^{-2}	−0.45	−1.77	$\phi_2^{i,g}$	1	−0.05	−2.88	$\phi_3^{i,c}$	10^{-2}	−0.04	−0.04				
$\sigma^{i,i}$	10^{-2}	0.18	10.57	$\phi_1^{g,s}$	10^{-2}	−0.04	−0.36	$\phi_2^{i,i}$	1	−0.03	−2.48	$\phi_1^{l,l}$	1	0.80	37.36				
$\sigma^{i,l}$	10^{-2}	0.27	4.01	$\phi_1^{g,c}$	10^{-2}	0.06	1.38	$\phi_2^{i,l}$	10^{-2}	0.39	0.23	$\phi_1^{l,s}$	1	0.02	1.32				
$\sigma^{i,s}$	10^{-2}	0.02	0.69	$\phi_1^{i,\pi}$	1	0.50	24.53	$\phi_2^{i,s}$	1	−0.07	−5.13	$\phi_1^{l,c}$	10^{-2}	0.24	0.35				
$\sigma^{i,c}$	10^{-2}	−0.02	−1.76	$\phi_1^{i,g}$	1	−0.13	−6.82	$\phi_2^{i,c}$	10^{-2}	0.52	0.47	$\phi_1^{s,l}$	1	0.24	9.34				
$\sigma^{l,l}$	10^{-2}	0.48	11.95	$\phi_1^{i,i}$	1	0.98	55.64	$\phi_3^{g^w,g^w}$	1	0.05	2.39	$\phi_1^{s,s}$	1	0.91	33.31				
$\sigma^{l,s}$	10^{-4}	−0.99	−0.38	$\phi_1^{i,l}$	1	−0.03	−1.88	$\phi_3^{g^w,p^o}$	1	−0.01	−1.25	$\phi_1^{s,c}$	10^{-2}	0.11	0.08				
$\sigma^{l,c}$	10^{-2}	0.01	1.27	$\phi_1^{i,s}$	1	0.09	6.05	$\phi_3^{g^w,\pi}$	1	0.02	4.99	$\phi_1^{c,l}$	1	−0.07	−4.32				
$\sigma^{s,s}$	10^{-2}	0.20	13.23	$\phi_1^{i,c}$	1	−0.01	−0.90	$\phi_3^{g^w,g}$	1	−0.02	−1.36	$\phi_1^{c,s}$	1	0.29	10.62				
$\sigma^{s,c}$	10^{-4}	−0.95	−1.04	$\phi_2^{g^w,g^w}$	1	−0.23	−13.63	$\phi_3^{g^w,i}$	10^{-2}	0.01	0.32	$\phi_1^{c,c}$	1	0.71	24.48				
$\sigma^{c,c}$	10^{-2}	0.10	17.34	$\phi_2^{g^w,p^o}$	1	−0.02	−1.60	$\phi_3^{g^w,l}$	10^{-2}	0.30	1.82								
σ^{π^*,π^*}	10^{-2}	0.02	5.85	$\phi_2^{g^w,\pi}$	1	−0.02	−4.33	$\phi_3^{g^w,s}$	10^{-2}	0.03	0.51								
σ^{r^*,r^*}	10^{-2}	0.20	3.52	$\phi_2^{g^w,g}$	1	0.02	1.25	$\phi_3^{g^w,c}$	10^{-2}	−0.09	−2.84								

Notes: The parameter names (columns 1, 4, 8, 12, and 16, respectively) closely follow the notation of Sections 2 and 3. Each parameter estimate (mode) is multiplied by the factor listed under the accompanying "times" column.

A.3.2 Stochastic trends and model fit

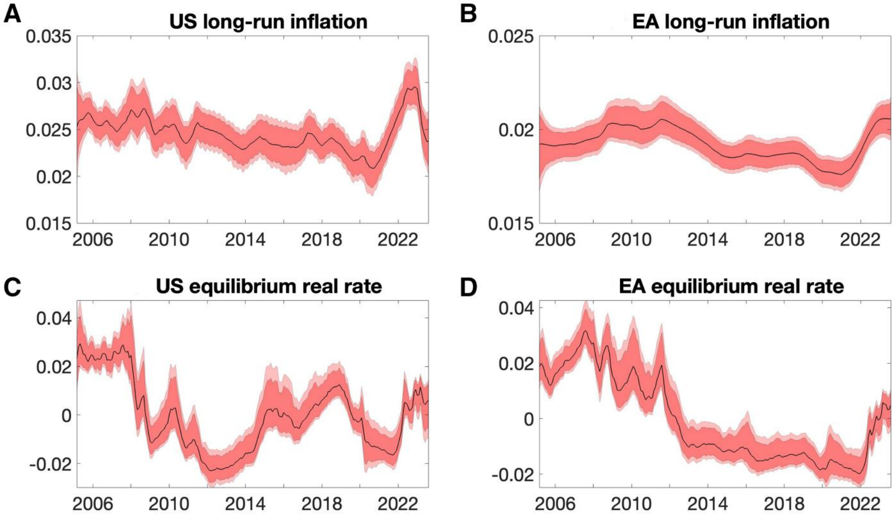

Figure A1. Long-run inflation and equilibrium real rate.

Notes: The top and bottom panels report the long-run inflation (real rate) for the US and EA, left and right panels, respectively. The estimations refer to a model with three lags.

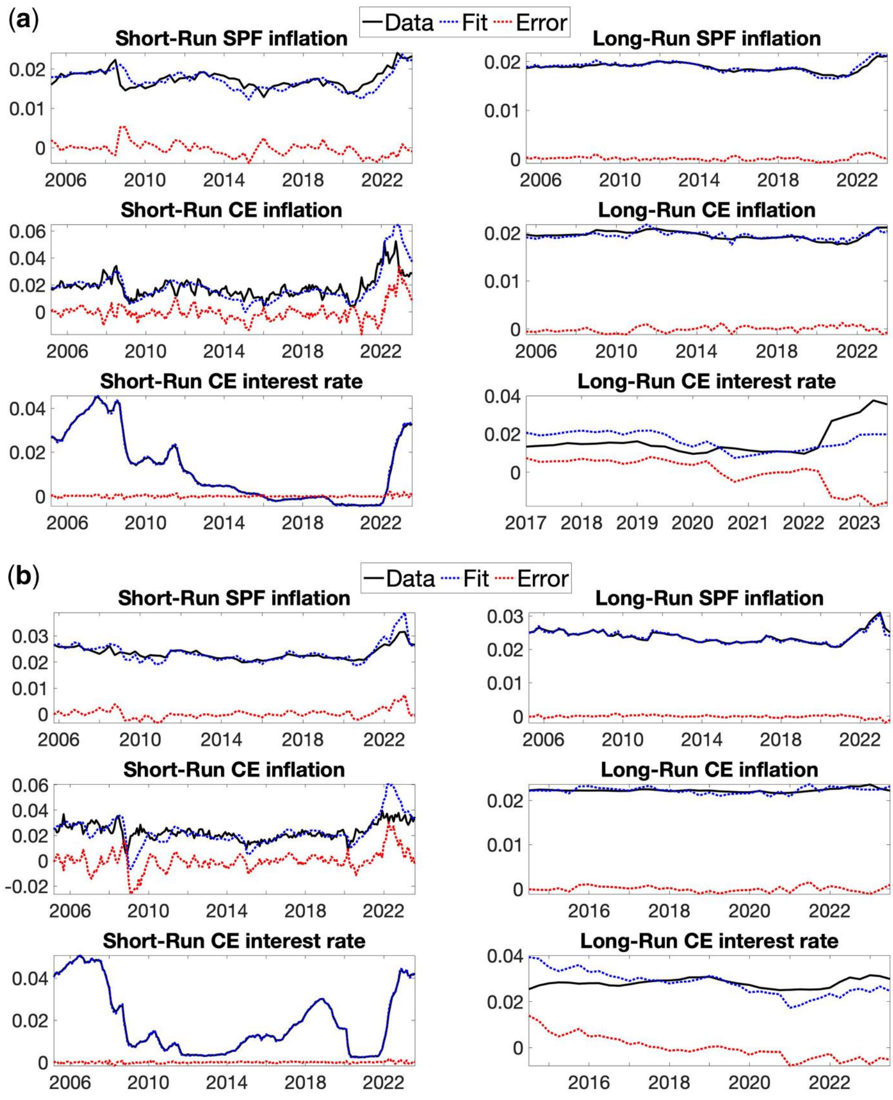

Figure A2. Fit of measurement equations.

Notes: Model fit is shown for the EA and US in subfigures (a) and (b), respectively. The top panels concern the short-term (left) and long-term (right) quarterly Survey of Professional Forecasters (SPF) forecasts of inflation. The middle panels report the fit of the Consensus Economics (CE) short-term (left) and long-term (right) inflation forecasts. The bottom panels plot the fit of the CE short-term (left) and long-term (right) interest rates forecasts. The observed data is in blue, the model fit is in black and the fitting error is in red.

A.4 Additional Tables

Table A6. EC and IRP summary statistics

	2-year					10-year			
Full sample	**EA EC**	**EA IRP**	**US EC**	**US IRP**	**Full Sample**	**EA EC**	**EA IRP**	**US EC**	**US IRP**
Min	0.0060	−0.0124	−0.0034	−0.0197	Min	0.0146	−0.0090	0.0150	−0.0057
Max	0.0459	0.0078	0.0313	0.0163	Max	0.0238	0.0085	0.0262	0.0123
Mean	0.0179	−0.0021	0.0170	0.0025	Mean	0.0176	0.0006	0.0178	0.0061
Std	0.0065	0.0041	0.0051	0.0054	Std	0.0016	0.0039	0.0020	0.0039
Autocorr	0.9374	0.8620	0.9523	0.8773	Autocorr	0.9551	0.9531	0.9422	0.9152
Pre Jan 2022	2-year				Pre Jan 2022	10-year			
Min	0.0060	−0.0124	−0.0034	−0.0197	Min	0.0146	−0.0090	0.0150	−0.0057
Max	0.0287	0.0058	0.0272	0.0163	Max	0.0196	0.0085	0.0209	0.0123
Mean	0.0164	−0.0024	0.0161	0.0022	Mean	0.0173	0.0003	0.0173	0.0063
Std	0.0037	0.0040	0.0043	0.0053	Std	0.0011	0.0039	0.0012	0.0039
Autocorr	0.8712	0.8872	0.9407	0.8732	Autocorr	0.9356	0.9572	0.9056	0.9223

Notes: The table reports some summary statistics of the decomposition of the 2-year (left panels) and 10-year (right panels) ILS rates in EC and IRP components. The top panels refer to the full sample, bottom panels refer to the sample that ends in January 2022.

A.5 Additional Figures

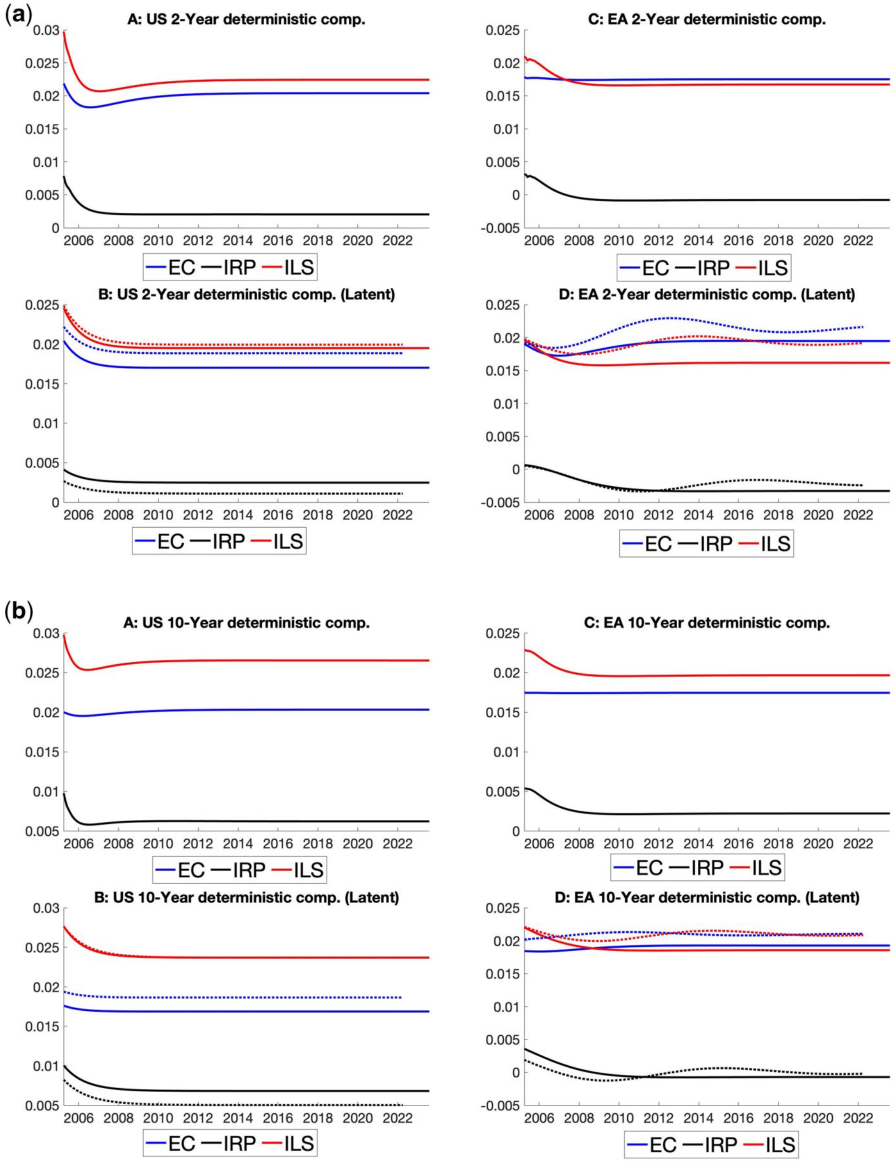

Figure A3. Two-year and 10-year rates: deterministic components.

Notes: (a) For the 2-year ILS rates, the subpanels of this figure report: (i) Panel A (E): the US (EA) ILS rates and their decompositions in EC and IRP; (ii) Panel B (F): the decomposition of the stochastic component of the US (EA) rates in EC and IRP; (iii) Panel C (G): the evolution of the deterministic component of the US (EA) rates and its decomposition in EC and IRP; (iv) Panel D (H): the evolution of the deterministic component of the US (EA) rates and its decomposition in EC and IRP for a model without unspanned risk factors and only Level, Slope, and Curvature. The dotted lines present the decomposition when estimating the latent model until March 2022. Panel (b) reproduces the same type of plots as (a), but for a 10-year ILS rates.

References

Aastveit, K. A., H. C. Bjørnland, and J. L. Cross. 2023. Inflation Expectations and the Pass-Through of Oil Prices. *Review of Economics and Statistics* 105: 733–743.

Abrahams, M., T. Adrian, R. Crump, E. Moench, and R. Yu. 2016. Decomposing Real and Nominal Yield Curves. *Journal of Monetary Economics* 84: 182–200.

Ang, A., G. Bekaert, and M. Wei. 2008. The Term Structure of Real Rates and Expected Inflation. *The Journal of Finance* 63: 797–849.

Ang, A, and M. Piazzesi. 2003. A no-Arbitrage Vector Autoregression of Term Structure Dynamics with Macroeconomic and Latent Variables. *Journal of Monetary Economics* 50: 745–787.

Bauer, M. D, and G. D. Rudebusch. 2017. Resolving the Spanning Puzzle in Macro-Finance Term Structure Models. *Review of Finance* 21: 511–553.

Bauer, M. D, and G. D. Rudebusch. 2020. Interest Rates under Falling Stars. *American Economic Review* 110: 1316–1354.

Berardi, A, and A. Plazzi. 2019. Inflation Risk Premia, Yield Volatility, and Macro Factors. *Journal of Financial Econometrics* 17: 397–431.

Baumann, U, M. Darracq Pariès, T. Westermann, M. Riggi, E. Bobeica, A. Meyler, B. Böninghausen, F. Fritzer, R. Trezzi, and J. Jonckheere. 2021. Inflation Expectations and their Role in Eurosystem Forecasting. Occasional Paper Series 264, European Central Bank.

Böninghausen, B., G. Kidd, and R. de Vincent-Humphreys. 2018. Interpreting Recent Developments in Market-Based Indicators of Longer-Term Inflation Expectations. *Economic Bulletin Articles* 6. European Central Bank.

Breach, T., S. D'Amico, and A. Orphanides. 2020. The Term Structure and Inflation Uncertainty. *Journal of Financial Economics* 138: 388–414.

Camba-Mendez, G, and T. Werner. 2017. The Inflation Risk Premium in the Post-Lehman Period. Working Paper Series 2033, European Central Bank.

Campbell, J. Y, and L. Viceira. 2001. Who Should Buy Long-Term Bonds? *American Economic Review* 91: 99–127.

Carriero, A., S. Mouabbi, and E. Vangelista. 2018. UK Term Structure Decompositions at the Zero Lower Bound. *Journal of Applied Econometrics* 33: 643–661.

Cecchetti, S, A. Grasso, and M. Pericoli. 2022. An Analysis of Objective Inflation Expectations and Inflation Risk Premia. Temi di discussione (Economic working papers) 1380, Bank of Italy, Economic Research and International Relations Area.

Chernov, M, and P. Mueller. 2012. The Term Structure of Inflation Expectations. *Journal of Financial Economics* 106: 367–394.

Christensen, J., J. Lopez, and G. Rudebusch. 2010. Inflation Expectations and Risk Premiums in an Arbitrage-Free Model of Nominal and Real Bond Yields. *Journal of Money, Credit and Banking* 42: 143–178.

Cieslak, A, and C. Pflueger. 2023. Inflation and Asset Returns. *Annual Review of Financial Economics* 15: 433–448.

Conflitti, C, and R. Cristadoro. 2018. Oil Prices and Inflation Expectations. Questioni di Economia e Finanza (Occasional Papers) 423, Bank of Italy, Economic Research and International Relations Area.

Dewachter, H, and L. Iania. 2011. An Extended Macro-Finance Model with Financial Factors. *Journal of Financial and Quantitative Analysis* 46: 1893–1916.

Dovern, J., U. Fritsche, and J. Slacalek. 2012. Disagreement among Forecasters in G7 Countries. *Review of Economics and Statistics* 94: 1081–1096.

Elliott, D., C. Jackson, M. Raczko, and M. Roberts-Sklar. 2015. Does Oil Drive Financial Market Measures of Inflation Expectations? Bank of England, Bank Underground.

Fan, J, and R. Li. 2001. Variable Selection via Nonconcave Penalized Likelihood and Its Oracle Properties. *Journal of the American Statistical Association* 96: 1348–1360.

Fisher, I. 1896. Appreciation and Interest: A Study of the Influence of Monetary Appreciation and Depreciation on the Rate of Interest with Applications to the Bimetallic Controversy and the Theory of Interest. *Publications of the American Economic Association* 11: 331–442.

Fleckenstein, M., F. A. Longstaff, and H. Lustig. 2017. Deflation Risk. *The Review of Financial Studies* 30: 2719–2760.

Galati, G., Z. Gorgi, R. Moessner, and C. Zhou. 2018. Deflation Risk in the Euro Area and Central Bank Credibility. *Economics Letters* 167: 124–126.

Galati, G., S. Poelhekke, and C. Zhou. 2011. Did the Crisis Affect Inflation Expectations? *International Journal of Central Banking* 7: 167–207.

Giannone, D., M. Lenza, and G. E. Primiceri. 2019. Priors for the Long Run. *Journal of the American Statistical Association* 114: 565–580.

Harvey, A. C. 1990. *Forecasting, Structural Time Series Models and the Kalman Filter*. Cambridge: Cambridge University Press.

Haubrich, J., G. Pennacchi, and P. Ritchken. 2012. Inflation Expectations, Real Rates, and Risk Premia: Evidence from Inflation Swaps. *Review of Financial Studies* 25: 1588–1629.

Holston, K., T. Laubach, and J. C. Williams. 2017. Measuring the Natural Rate of Interest: International Trends and Determinants. *Journal of International Economics* 108: S59–S75.

Hördahl, P, and O. Tristani. 2014. Inflation Risk Premia in the Euro Area and the United States. *International Journal of Central Banking* 10: 1–47.

Höynck, C, and L. Rossi. 2023. The Drivers of Market-Based Inflation Expectations in the Euro Area and in the US. *Economics Letters* 232: 111323.

Joslin, S., A. Le, and K. J. Singleton. 2013. JFEC Invited Paper: Gaussian Macro-Finance Term Structure Models with Lags. *Journal of Financial Econometrics* 11: 581–609.

Joslin, S., K. Singleton, and H. Zhu. 2011. A New Perspective on Gaussian Dynamic Term Structure Models. *Review of Financial Studies* 24: 926–970.

Joyce, M. A., P. Lildholdt, and S. Sorensen. 2010. Extracting Inflation Expectations and Inflation Risk Premia from the Term Structure: A Joint Model of the UK Nominal and Real Yield Curves. *Journal of Banking & Finance* 34: 281–294.

Kilian, L. 2009. Not All Oil Price Shocks Are Alike: Disentangling Demand and Supply Shocks in the Crude Oil Market. *American Economic Review* 99: 1053–1069.

Kilian, L, and X. Zhou. 2022. The Impact of Rising Oil Prices on U.S. inflation and Inflation Expectations in 2020–23. *Energy Economics* 113: 106228.

Kozicki, S, and P. Tinsley. 2001. Shifting Endpoints in the Term Structure of Interest Rates. *Journal of Monetary Economics* 47: 613–652.

Liu, Y, and J. C. Wu. 2021. Reconstructing the Yield Curve. *Journal of Financial Economics* 142: 1395–1425.

Mouabbi, S., J.-P. Renne, and Tschopp. 2023. *The Dynamic Nature of Macroeconomic Risk*. Mimeo.

Neri, S., G. Bulligan, S. Cecchetti, F. Corsello, A. Papetti, M. Riggi, C. Rondinelli, and A. Tagliabracci. 2022. "On the Anchoring of Inflation Expectations in the Euro Area." Questioni di Economia e Finanza (Occasional Papers) 712, Bank of Italy, Economic Research and International Relations Area.

Sims, C. A. 2000. Using a Likelihood Perspective to Sharpen Econometric Discourse: Three Examples. *Journal of Econometrics* 95: 443–462.

Stevens, A, and J. Wauters. 2021. Is Euro Area Lowflation Here to Stay? Insights from a Time-Varying Parameter Model with Survey Data. *Journal of Applied Econometrics* 36: 566–586.

Stoffer, D. S, and K. D. Wall. 1991. Bootstrapping State-Space Models: Gaussian Maximum Likelihood Estimation and the Kalman Filter. *Journal of the American Statistical Association* 86: 1024–1033.

Tibshirani, R. 1996. Regression Shrinkage and Selection Via the Lasso. *Journal of the Royal Statistical Society: Series B (Methodological)* 58: 267–288.

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